Chapter 4. Indications for Mechanical Ventilation

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Indications for Mechanical Ventilation: Introduction

This chapter discusses the indications for mechanical ventilation in adult patients. We focus on patients who are already in the intensive care unit (ICU) or who are considered for transfer to the ICU, that is, patients with new onset of signs and symptoms over minutes or hours. We do not deal with the indications for mechanical ventilation for chronic respiratory failure or in pediatric patients; these subjects are covered in Chapters 23, 18, 33, and 34.

There is a paucity of research—and no clinical trials—on the indications for mechanical ventilation. This situation contrasts with the growing amount of research on the discontinuation of mechanical ventilation. Although it is tempting to apply indices used for predicting the outcome of weaning trials as indices to identify patients who require mechanical ventilation, such an approach has not been tested. It is also probably unwise.

Two factors account for the limited research on indications for mechanical ventilation. First, such patients are extremely ill. Any intervention—such as careful collection of physiologic measurements—that delays institution of ventilation might be viewed as unethical. Second, the nosology of respiratory failure is unsatisfactory (see "Nosology" below). In everyday practice, clinicians do not decide to institute mechanical ventilation because a patient meets certain diagnostic criteria. Instead, clinicians typically decide to institute ventilation based on their assessment of a patient's signs and symptoms. This decision is also grounded on a foundation of solid biomedical theory, specifically principles of pulmonary pathophysiology. Accordingly, we develop our discussion of ventilator indications along these two lines: physical examination and pathophysiologic principles.

Overall Assessment

Clinical presentations that cause a physician to institute mechanical ventilation are protean. They range from patients presenting with frank apnea to patients with clinical signs of increased work of breathing with or without laboratory evidence of impaired gas exchange.[1]

Apnea

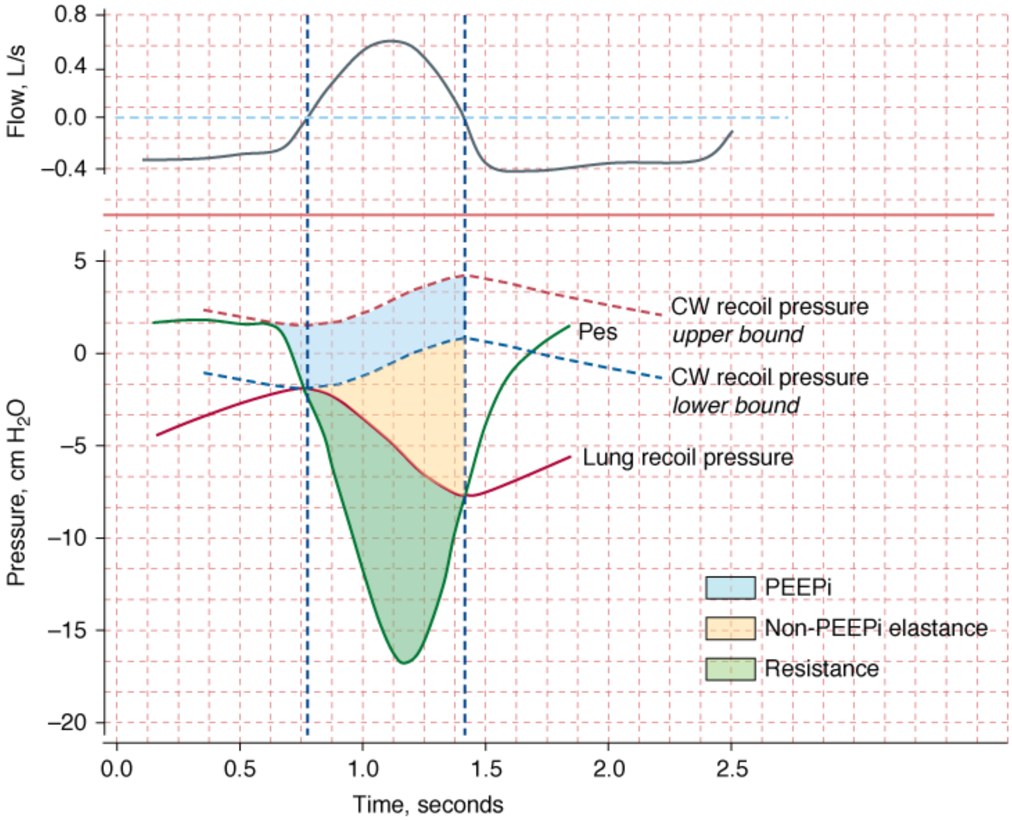
Apneic patients, such as those who have suffered catastrophic central nervous system (CNS) damage, need immediate institution of mechanical ventilation. To advocate controlled trials to determine the need for mechanical ventilations in apneic patients is unethical.

Clinical Signs of Increased Work of Breathing

Asthma, chronic obstructive pulmonary disease (COPD), pneumonia, cardiogenic pulmonary edema, and acute respiratory distress syndrome (ARDS) are just a few of the many conditions that cause an increase in work of breathing and, with it, increased energy expenditure by the respiratory muscles.

The energy expenditure of the respiratory muscles can be quantified in terms of pressure-time product[2]—the time integral of the difference between the esophageal pressure tracing and the estimated recoil pressure of the chest wall[3,4] (Fig. 4-1). The pressure-time product of patients in acute respiratory failure is about four times[5–7] the normal value (100 cm H_2O·s/min), and it can be increased sixfold in individual patients.[5,6] The inspiratory pressure-time product can be partitioned into resistive, elastic, and intrinsic positive end-expiratory pressure (PEEP) components (Fig. 4-1).[6] Patients in respiratory distress typically have a 30% to 50% greater inspiratory resistance,[6] 100% greater dynamic elastance,[6] and 100% to 200% greater intrinsic PEEP[5,6] than do similar patients who are not in acute respiratory failure. Inspiratory effort is almost equally divided in offsetting intrinsic PEEP, elastic recoil, and inspiratory resistance.[6] The increase in respiratory effort means that the respiratory muscles account for a much larger fraction of the body's oxygen consumption. In healthy subjects, this fraction is only 1% to 3% of total oxygen consumption. In patients with acute hypoxemic respiratory failure and shock who are undergoing cardiopulmonary resuscitation, the respiratory muscles account for approximately 20% of total oxygen consumption.[8]

Figure 4-1



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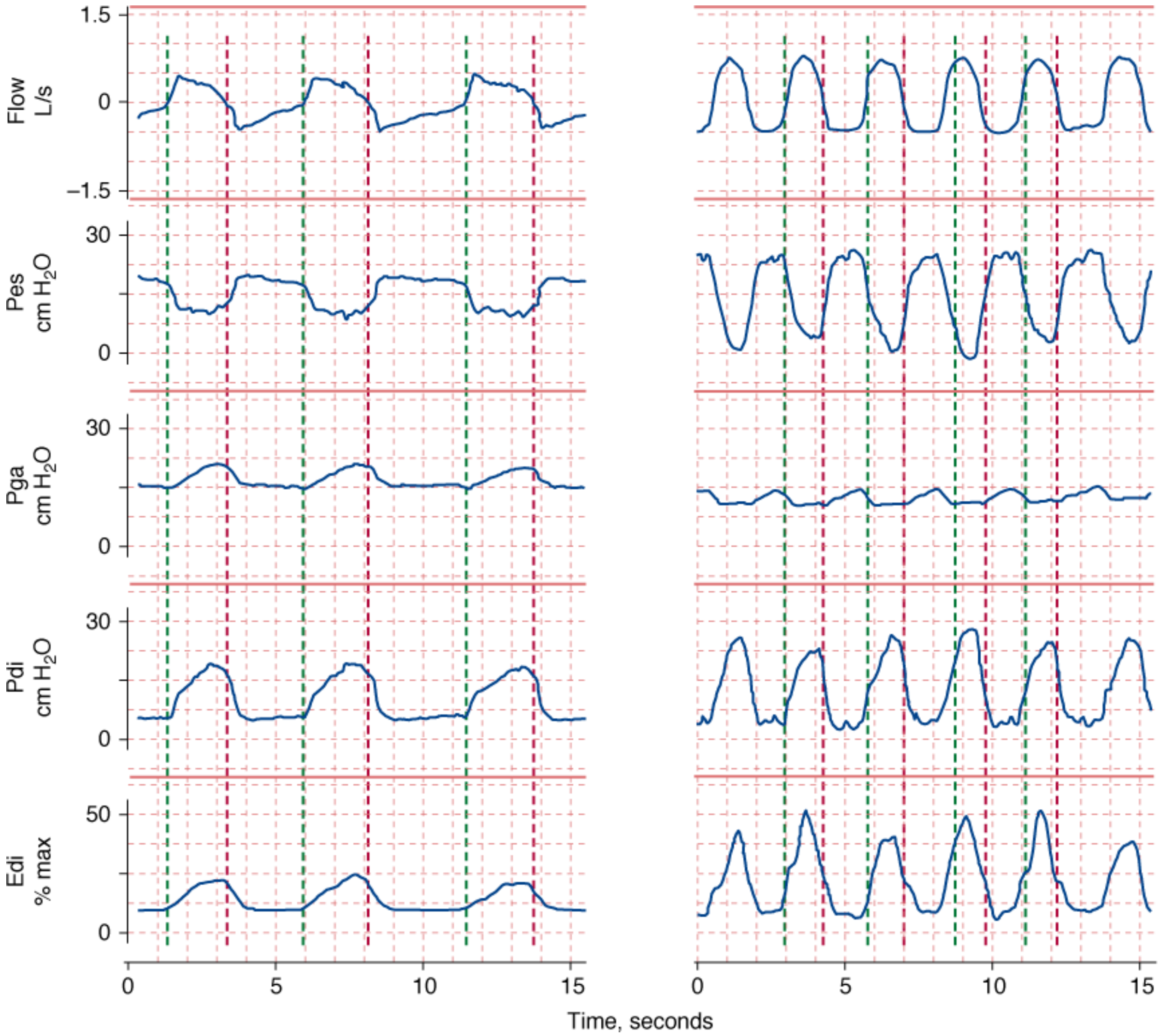
Flow (*inspiration upward*) and pressure tracings during spontaneous breathing. Recoil pressures of the chest wall (*CW*) and lung are calculated from dynamic elastances of the chest wall and lung, respectively, and lung volume. Inspiratory pressure-time product (PTP) is calculated using the integral of the difference between esophageal pressure (Pes) and CW recoil pressure from the onset of the rapid decrease in Pes to the transition from inspiratory to expiratory flow (upper-bound PTP). The component of PTP caused by intrinsic positive end-expiratory pressure (*PEEPi*) is computed using the integral of the difference between the upper-bound PTP and CW recoil pressure from the onset of rapid decrease in Pes to the transition from inspiratory to expiratory flow (lower-bound PTP). The component of PTP caused by non-PEEPi elastance is computed using the integral of the difference between lung recoil pressure and lower-bound CW recoil pressure from the onset of inspiratory flow to the moment of transition from inspiratory to expiratory flow. The resistive fraction of PTP is computed using the integral of the difference between Pes and lung recoil pressure. The *vertical interrupted lines* represent points for zero flow. (Modified, with permission, from Jubran and Tobin.[6])

Increased work by the respiratory muscles causes respiratory distress. Clinical manifestations of respiratory distress include nasal flaring, retraction of the eyelids, accessory muscle recruitment, expiratory muscle recruitment (Fig. 4-2), tracheal tug, intercostal recession, tachypnea, tachycardia, hypertension or hypotension, diaphoresis, and changes in mental status.

Figure 4-2



Respiratory effort during unassisted respiration. Recordings of flow (*inspiration upward*), esophageal (*Pes*), gastric (*Pga*), and transdiaphragmatic (*Pdi*) pressures and electrical activity of the diaphragm (*Edi*) in a stable patient with COPD (*left*) and in a patient with respiratory failure (*right*). The *green vertical lines* indicate the onset of inspiratory flow and the *red vertical lines* indicate the onset of expiratory flow. The excursions in Pes and Edi in the patient in respiratory failure are three times greater than in the stable patient, signifying heightened respiratory motor output. The increase in Pga during exhalation in the patient with respiratory failure signifies expiratory muscle recruitment.

Facial Signs of Respiratory Distress

Many intensivists decide to institute mechanical ventilation based on a patient's facial appearance.[9] Gilston has provided an insightful description of many signs that go unstated in reviews on mechanical ventilation.[9] Consider, for example, the mouth. At an early stage of respiratory distress, the mouth opens slightly and to a variable extent during inhalation (Fig. 4-3). At a later stage, the mouth opens throughout the respiratory cycle. Patients may switch to mouth breathing perhaps to decrease respiratory work[9,10] and physiologic dead space ventilation.[9] An open mouth is sometimes seen in patients with a tracheostomy (Fig. 4-4) and in patients receiving ventilator support.[9] The tongue may be seen to jerk in unison with inspiratory efforts.[9]

Figure 4-3

Change in facial appearance during the development and resolution of acute respiratory failure resulting from congestive heart failure and exacerbation of chronic obstructive pulmonary disease. *Left upper panel:* The patient is dyspneic and her mouth is open on inhalation. *Right upper panel:* The patient exhibits pursed-lip breathing on exhalation. Over the ensuing 24 hours, the patient developed hypercapnic respiratory failure and failed a trial of noninvasive ventilation (not shown). *Left lower panel:* The patient is intubated and receiving mechanical ventilation. *Right lower panel:* The patient is successfully extubated 4 days after institution of mechanical ventilation.

Figure 4-4

Change in the configuration of the mouth in a patient with a tracheostomy who becomes dyspneic. *Left:* The patient is resting during full ventilator support and his mouth is closed. *Middle:* Twelve minutes after disconnection from the ventilator, the patient has developed dyspnea and anxiety and his mouth is open. *Right:* Thirty minutes after reconnection to the ventilator, the patient's respiratory distress has resolved and his mouth is closed.

Some distressed patients also exhibit pursed-lip breathing during exhalation (see Fig. 4-3).[9] In stable, ambulatory patients with COPD, pursed-lip breathing is associated with an increase in tidal volume,[11] a decrease in respiratory rate,[11] and, in patients with severe obstruction, with a decrease end-expiratory lung volume.[12] Pursed-lip breathing can improve the arterial tensions of both carbon dioxide (Pa_{CO_2}) and oxygen (Pa_{O_2}), whereas oxygen uptake ($\dot{V}_{O_2}$) remains unchanged.[11] The latter finding suggests that pursed-lip breathing may allow a decrease in cardiac output without changing tissue oxygenation.[11] Alternatively, if cardiac output does not decrease, pursed-lip breathing may increase mixed venous oxygen saturation, resulting in better tissue oxygenation.[11] Pursed-lip breathing is thought to improve gas exchange by preventing airway collapse. As a result, gas trapping is decreased, resulting in an increase in tidal volume.[11]

A few patients in respiratory distress moan during exhalation. Such moans have been compared with the grunting that is typical of neonates with the respiratory distress syndrome.[9] Grunting results from glottic closure and expiratory muscle recruitment during early exhalation.[13] It is associated with a rise in transpulmonary pressure and oxygenation.[13] If grunting is prevented by tracheal intubation, oxygenation deteriorates.[13] Use of continuous positive airway pressure (CPAP) improves oxygenation and eliminates grunting.[14] The improvement in oxygenation may result from grunting acting as a natural form of PEEP that recruits collapsed alveoli and partially overcomes inequalities in gas distribution caused by differing time constants. Of course, grunting also can arise with disease outside the thorax, such as with an acute abdomen.[15]

Nasal flaring, another facial sign of respiratory distress, is caused by contraction of the alae nasi, the dilator muscle of the external nares.[10] In adults, nasal flaring reduces nasal resistance by approximately 40% to 50% and total airway resistance by approximately 10% to 30%.[10] Factors regulating alae nasi activity include chemical stimuli that cause hyperpnea (hypoxia and hypercapnia),[10,16,17] inspiratory resistive loading,[17] and local stimuli (negative intraluminal nasal pressure).[16] The proportion of patients in respiratory distress who present with nasal flaring is unknown, as is the level of interobserver agreement in detecting flaring. Ventilator support reduces or eliminates alae nasi activity.[18,19]

Diaphoresis, often best detected on the forehead,[9] accompanies respiratory distress in some patients. Among forty-nine patients admitted to the emergency ward for acute bronchial asthma, Brenner et al[20] found that nine patients had profuse sweating. This subgroup displayed greater abnormalities in peak expiratory flow rate and Pa_{CO_2}. In patients with respiratory distress, diaphoresis may result from increased work of breathing,[21] sympathetic stimulation,[21,22] and hypercapnia-associated cutaneous vasodilation.[21,23] In contrast, diaphoresis in patients with heart failure often is associated with hypoperfusion of the skin, vasoconstriction, and cold extremities.[21]

Mentation can be evaluated by inspection of the face and by simple questioning. With early respiratory distress, nearly all patients are anxious, and their eyelids are retracted. As distress increases, the level of consciousness often decreases, and the lids tend to fall (Fig. 4-5). Instead of remaining alert to their surroundings, patients gaze vacantly ahead.[9] If respiratory failure is left untreated, apathy leads to drowsiness and then coma. These changes in mentation arise because of the underlying cause of respiratory failure (decreased cardiac output in cases of shock, impaired neurologic function in sepsis), acute hypercapnia,[24] or to a lesser extent, hypoxemia.[24,25] In a classic description, Campbell noted that most (nonhypotensive) patients with an exacerbation of COPD have preserved consciousness on arrival to the emergency room despite Pa_{O_2} being as low as 20 to 40 mm Hg.[25] Although extremely useful in overall patient assessment, facial signs of respiratory distress do not necessarily translate into an automatic decision to intubate a patient (Fig. 4-5).

Figure 4-5

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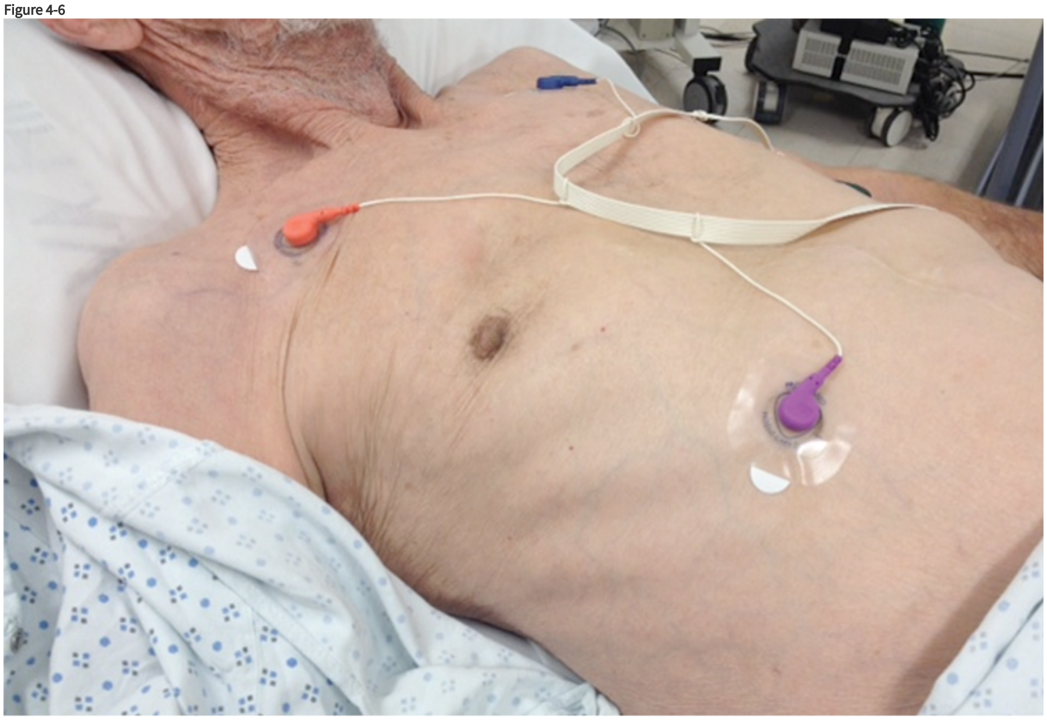
Drooping of the eyelids accompanying deterioration of mental status during an episode of respiratory failure. *Left:* A patient developed respiratory compromise secondary to atrial fibrillation and heart failure 2 days after prostate surgery. The patient is drowsy with drooping eyelids, and is moderately dyspneic with an open mouth and groans during exhalation. *Middle:* Despite respiratory compromise, the patient is able to drink. *Right:* After 2 days of medical therapy, the patient's dyspnea has resolved and he is alert, cheerful, and talkative.

Accessory and Expiratory Muscle Recruitment

Increased respiratory loads in healthy subjects[26,27] and in ambulatory patients with COPD[28] are met with a proportionately greater use of the rib-cage muscles than of the diaphragm.[26–28] As the load increases, the expiratory muscles are recruited.[26,27,29] In addition to increased activity of rib cage and abdominal muscles, the respiratory centers may increase activity of the accessory muscles, especially the sternomastoids.[30–32] The sternomastoids are activated in patients with respiratory compromise[30] and in healthy subjects breathing with a high level of inspiratory effort. The threshold for sternomastoid activation, however, is lower in patients.[31] In patients with respiratory distress,[5,33] sternomastoid recruitment can be phasic (during inhalation) (Fig. 4-6) or tonic.[29,32] Some patients hold their head off the pillow to enhance sternomastoid action.[9]

Figure 4-6



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Sternomastoid muscle recruitment and intercostal recession during an episode of respiratory distress.

Tachypnea, Paradox, Tracheal Tug, Intercostal Recessions

Changes in respiratory rate are one of the most useful signs in evaluating the need for mechanical ventilation. Tachypnea is a near-universal sign accompanying respiratory distress.

Obtaining reliable measurements of respiratory rate and interpreting the values are not straightforward. First, bedside assessment often is inaccurate. In one study, 40% of nurses' estimations of respiratory frequencies deviated by more than 20% from the true value.[34] Agreement as to the presence of tachypnea among physicians, expressed as a κ value (κ of 0 indicates that agreement is no better than chance; κ of 1 indicates complete agreement), was only 0.25.[35] Second, the within-day (within an individual) coefficient of variation in respiratory rate among young, healthy adults is 21 ± 12%; in old, healthy adults, it is 29 ± 11%.[36] Thus, accurate quantification of respiratory rate requires counting more breaths than contained in the usual 15-second sample.[36,37] Third, the day-to-day coefficient of variation of respiratory rate is 7 ± 2% in healthy individuals[36]; the value in patients with pulmonary diseases is unknown. Fourth, the typical respiratory rate in patients with different disease states varies from one state to the next[37] (Table 4-1); a rate that is judged high in a previously healthy subject may arouse no concern in a patient with restrictive disease. The upper limit of normal (mean + 2 standard deviations [SD]) in health is 22 breaths/min.[37] The equivalent value for stable patients with COPD is 30 breaths/min, and for patients with restrictive lung disease, 44 breaths/min.[37]

Table 4-1: Respiratory Rates in Health and Disease

Condition	Number of Subjects	Mean (breaths/min)	SD	Mean + 2 SD
Healthy nonsmoker	65	16.6	2.8	22.2
Healthy smoker	22	18.3	3.0	24.3
Asthma	17	16.6	3.4	23.4
COPD, eucapnia	16	20.4	4.1	28.6
COPD, hypercapnia	12	23.3	3.3	29.9
Restrictive lung disease	14	27.9	7.9	43.7
Pulmonary hypertension	7	25.1	6.4	37.9
Chronic anxiety	13	18.3	2.8	23.9

Abbreviations: COPD, chronic obstructive pulmonary disease; SD, standard deviation.

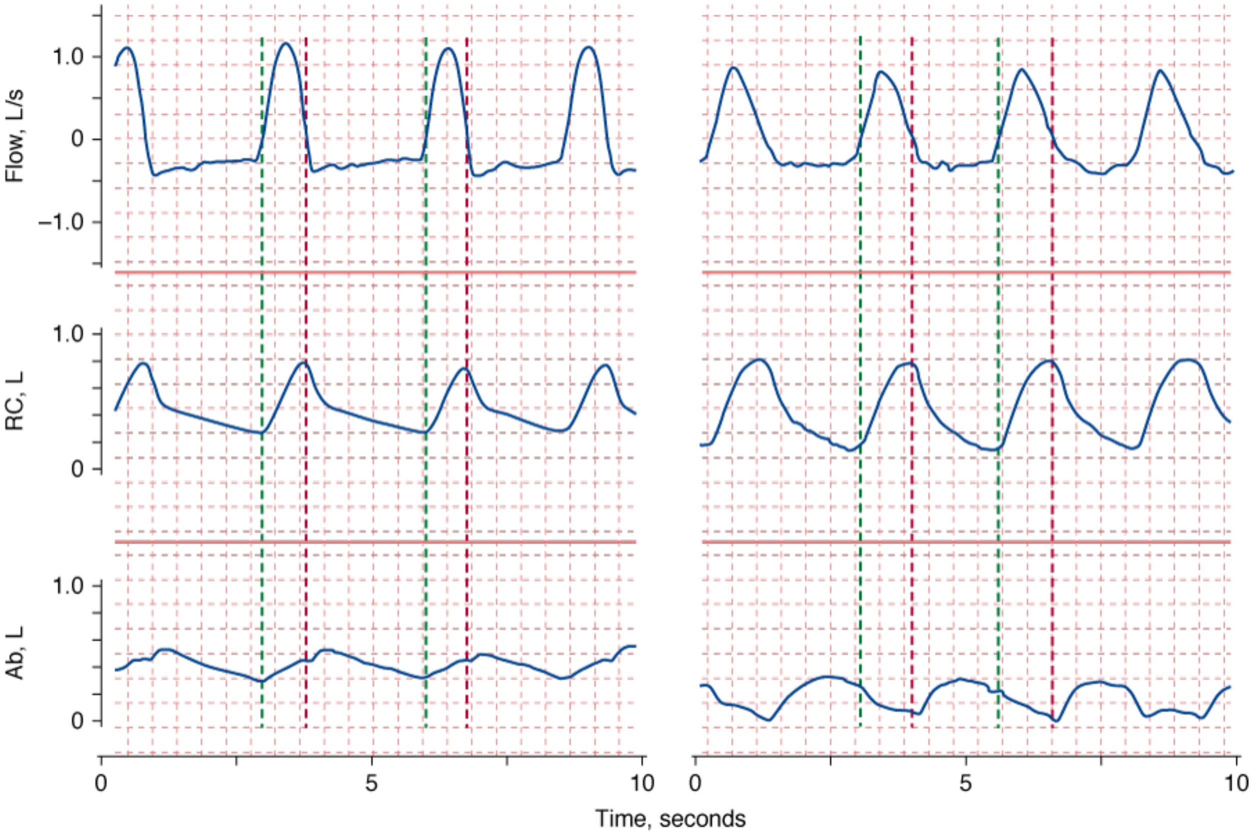
Source: Data from Tobin et al.[37]

Despite limitations in its measurement, tachypnea is an important clinical sign. In a retrospective case-controlled study of patients discharged from an ICU, the only continuous variables that predicted readmission to the ICU were higher respiratory rate (24 vs. 21 breaths/min) and lower hematocrit.[38] Readmitted patients had a much higher mortality than the control patients, 42% and 7%, respectively, and respiratory problems accounted for more than half the readmissions. Of eighteen patients who were discharged from the ICU with a respiratory rate of more than 30 breaths/min, twelve required readmission. In a study of patients who had experienced a cardiopulmonary arrest, 53% had documented deterioration in respiratory function in the 8 hours preceding the arrest.[39] Of interest, respiratory rate was elevated in most patients (mean: 29 ± 1 [SE (standard error)] breaths/min), whereas other routine laboratory tests showed no consistent abnormalities. That detection of tachypnea did not lead to a change in patient management (in an effort to prevent the arrest) led the authors to surmise that physicians do not fully appreciate its clinical importance.

Shallow respiration (when measured with instrumentation) is common in patients with acute respiratory distress.[40] Judging tidal volume as shallow based on physical examination is very unreliable.[41,42] Clinical skill in this task is not improved by years of experience.[42]

Patients in distress commonly display abnormal chest wall movements.[43,44] Abnormal movements can be separated into three categories. One, asynchrony, consists of a difference in the rate of motion of the rib cage and abdomen (Fig. 4-7). Two, paradox, consists of one compartment moving in the opposite direction to tidal volume (Fig. 4-7). The third abnormality is greater-than-normal breath-to-breath variation in the relative contribution of the rib cage and abdomen to tidal volume; this pattern, termed *respiratory alternans,* represents recruitment and derecruitment of the accessory intercostal muscles and the diaphragm. In the past, it was thought that these three abnormalities of motion represented respiratory muscle fatigue.[45] It is now known that they represent signs of increased load and occur in the absence of fatigue.[46] These abnormalities are seen not only in patients with respiratory distress,[47] but also in some ambulatory patients[37,47] and during sleep (sleep apnea syndrome).[48]

Figure 4-7



Recordings of flow (*inspiration upward*), rib cage (*RC*), and abdominal (*Ab*) cross-sectional areas in two patients in respiratory distress. The *green vertical lines* indicate the onset of inspiratory flow and the *red vertical lines* indicate the onset of expiratory flow. On the *left,* expansion of the rib cage is occurring faster than expansion of the abdomen (asynchrony). On the *right,* while the rib cage expands during inspiration, the abdominal cross-sectional area is getting smaller (paradox).

Increased tidal swings in intrathoracic pressure are axiomatic to increases in the work of breathing. The greater downward movement of the diaphragm tends to pull down the trachea (just as a sexton ringing a bell) with each inspiration,[49] producing a sign termed *tracheal tug.* Tracheal tug correlates closely with severity of airway obstruction (Fig. 4-8).[50]

Figure 4-8

Tracheal tug in acute respiratory failure. *Left:* During exhalation, the cricoid cartilage is located 2 fingerbreaths above the suprasternal notch (normally, at least 4 fingers). *Right:* During inhalation, the cricoid cartilage is pulled below the suprasternal notch and the thyroid cartilage is 1 fingerbreath above the suprasternal notch, whereas it was 5 fingerbreaths above the notch on exhalation (*left*).

The intercostal spaces normally bulge inward during inhalation and outward during exhalation.[15] Inspiratory retraction of the intercostal space—serving as a window into the pleural space—is increased in patients with respiratory disease (see Fig. 4-6). The suprasternal fossa also moves inward in direct proportion to swings in pleural pressure.[51,52] Focal exaggerated retraction of the intercostal space can occur with a flail chest. Focal expiratory bulging may be seen on the side of a tension pneumothorax or over the area of a flail chest.[15] As with other physical signs, studies often reveal poor agreement among physicians.[35,53] For example, Godfrey et al[53] found agreement among eleven relatively experienced chest physicians in identifying tracheal tug to be midway between chance and maximum possible agreement. Spiteri et al[35] found poor agreement among experienced physicians for detecting reduced chest movements ($\kappa = 0.38$) and cricosternal distance ($\kappa = 0.28$)[35]; they did not address tracheal tug or inspiratory retractions.

The interpretation of data generated by studies that quantify physical signs is hazardous. The entity that researchers are quantifying can be very different from the skill involved in the physician's actions. Physical examination is an art—learned through apprenticeship, not out of a book. Thus, we should bear in mind Braque's caution about art appreciation: "The only thing that matters in art can't be explained." Likewise, the essence of physical examination is its tacit coefficient; the explicit, measurable components may be the least relevant. Another problem with research on physical signs is test-referral bias. For example, the studies of Godfrey et al[53] and Spiteri et al[35] were confined to patients with respiratory diseases; a more appropriate design also would have included healthy subjects and patients with diseases not affecting the lungs. These flaws in the methodology of such studies markedly underestimate the diagnostic power of physical examination.

Cardiovascular Signs of Respiratory Distress

Respiratory distress frequently is associated with tachycardia and hypertension. Tachycardia and hypertension likely are caused by increased sympathetic discharge.[54] In some patients, such as those with sepsis, cardiac impairment, or severe hypoxemia, respiratory distress is associated with hypotension and not hypertension.

Pulsus paradoxus is defined as an inspiratory fall in systolic pressure of greater than 10 mm Hg.[15] Pulsus paradoxus is very common in patients with an exacerbation of asthma but also in patients with COPD, shock, and pericardial tamponade (see Chapter 36).[15,20]

Nonuniform Presentation

Patients with impending respiratory failure do not have a uniform presentation. The spectrum ranges from a patient complaining of dyspnea to a patient with impending respiratory arrest. Several factors are responsible. Patients differ in the balance between work of breathing and the capacity of the respiratory muscles to generate pressure. They also differ in the central processing of neural afferents. For instance, patients with a history of near-fatal asthma have a blunted perception of dyspnea,[55] reduced sensitivity to added inspiratory resistive loads,[56] and a reduced chemosensitivity to hypoxia.[55] Alexithymia, the difficulty in perceiving and expressing emotions and body sensations, occurs more often in patients who experience near-fatal asthma than in less-severely affected patients.[57] In addition, hypoxia, and possibly hypercapnia, can impair sensations of respiratory load.[58]

Impending Respiratory Failure

Development of impending respiratory failure is a commonly listed indication for mechanical ventilation.[59] But *impending respiratory failure* has no clear definition. Some clinicians use the term to mean development of severe tachypnea, diaphoresis, and use of accessory muscles of respiration; others use it to mean agonal breathing.

In some circumstances, physicians do not institute mechanical ventilation until they obtain results of diagnostic testing, such as chest radiographs, electrocardiograms, or arterial blood-gas analyses. Even in this situation, clinicians commonly do not change their mind when the test results are different from those expected. In most circumstances, a patient's clinical presentation is so dramatic that mechanical ventilation is instituted without performing arterial blood-gas analysis. If arterial blood-gas results are not available before connecting a patient to a ventilator, they are almost invariably available shortly after. Arterial blood-gas analysis is helpful in choosing the type of support best suited to a patient's needs. Analysis also serves to classify patients into two broad groups: hypoxemic respiratory failure (Table 4-2) and hypercapnic respiratory failure; some patients display features of both.[60]

Table 4-2: Common Causes of Hypoxemic Respiratory Failure

Pneumonia
Cardiogenic pulmonary edema
Acute respiratory distress syndrome
Aspiration of gastric contents
Multiple trauma
Immunocompromised host with pulmonary infiltrates
Pulmonary embolism

Hypoxemic Respiratory Failure

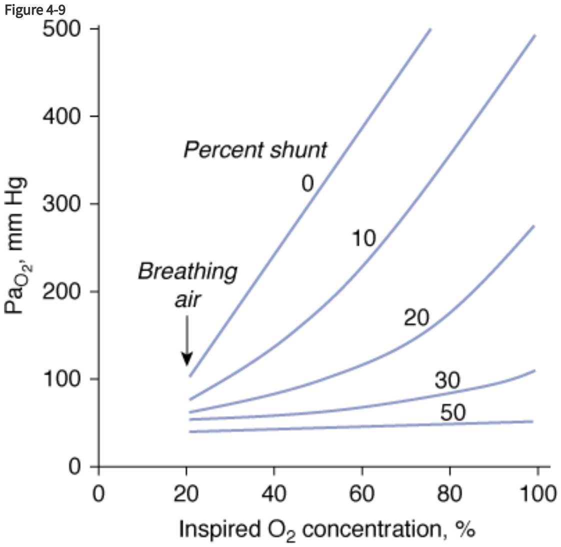
Pathophysiology

The pathophysiologic mechanisms responsible for hypoxemia can be grouped into two broad categories depending on whether there is (or is not) an increased alveolar-arterial oxygen gradient (A-aD_{O_2}).* An increased A-aD_{O_2} results from either ventilation–perfusion ($\dot{V}_A/\dot{Q}$) abnormalities or excessive right-to-left shunt. (Diffusion impairment, a third cause of increased A-aD_{O_2}, plays only a marginal role in the development of hypoxemia.) Patients with hypoxemic respiratory failure and a normal A-aD_{O_2} typically have alveolar hypoventilation or inadequate inspiratory partial pressure of oxygen. Hypoxemia results from a low inspired P_{O_2} or when the fractional concentration of inspired oxygen (FI_{O_2}) is less than 0.21, such as at high altitude or when O_2 is consumed from ambient gas secondary to a fire. A low also can arise during anesthesia if a low-oxygen gas mixture is administered inadvertently. Hypoxemic respiratory failure also can be caused by the combination of a decreased mixed venous oxygen content and impaired gas exchange, such as in patients with heart failure and concurrent $\dot{V}_A/\dot{Q}$ derangements or increased shunt.

A right-to-left shunt is present when venous blood returning from the tissues passes to the systemic arterial circulation without coming into contact with gas-containing alveoli. The shunt is the major mechanism of abnormal gas exchange in patients with pulmonary edema, ARDS, pneumonia, and atelectasis. In all these instances, the shunt results from the perfusion of alveoli that are unventilated because they are filled with fluid or collapsed.

A characteristics feature of a shunt is the failure of Pa_{O_2} to increase to the expected level when a patient breathes 100% oxygen (Fig. 4-9).[61]

Figure 4-9



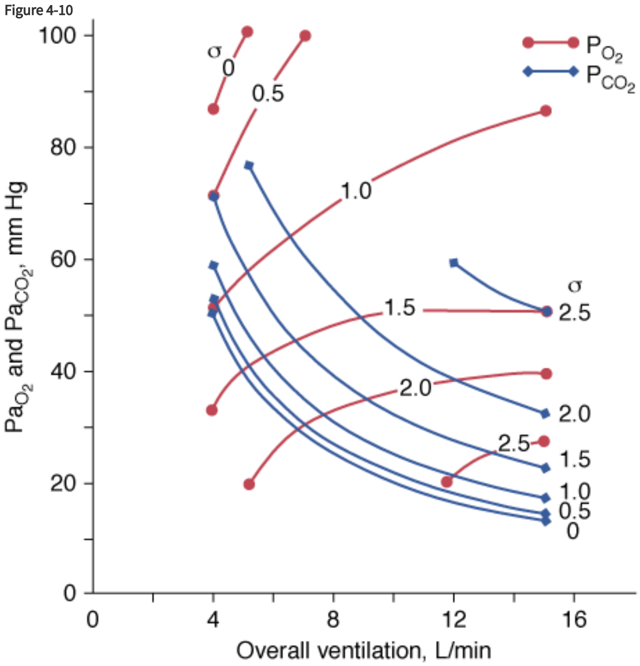
Relationship between arterial P_{O_2} (Pa_{O_2}) and increases in inhaled oxygen concentrations for different levels of shunt. When the shunt fraction is 30% or more of cardiac output, Pa_{O_2} increases little despite marked increases in inhaled oxygen concentration. The plot is a simplification that ignores factors such as cardiac output and oxygen uptake, which influence the location of the lines. (Modified, with permission, from West.[61])

In patients with a large shunt, Pa_{CO_2} may be low because the low Pa_{O_2} stimulates respiratory motor output and minute ventilation ($\dot{V}_E$).[62] A shunt typically produces more severe hypoxemia than does $\dot{V}_A/\dot{Q}$ inequality.

Optimal uptake of oxygen depends on proper matching of ventilation and perfusion within the lung. In semirecumbent young healthy subjects breathing room air, the range (or dispersion) of $\dot{V}_A/\dot{Q}$ ratios is quite small: More than 95% of both ventilation and perfusion is limited between $\dot{V}_A/\dot{Q}$ ratios of 0.3 and 2.1.[63] The dispersion increases with age.[63] With pulmonary disease, the range of $\dot{V}_A/\dot{Q}$ ratios widens,[64] varying from 0 (perfused but unventilated, i.e., shunt) to infinity (ventilated by unperfused, i.e., alveolar dead space). In other words, $\dot{V}_A/\dot{Q}$ inequality does not refer to alterations in the ratio of total ventilation to total perfusion, which constitute global hyperventilation or hypoventilation. (One lung could receive all ventilation and the other all perfusion for an overall $\dot{V}_A/\dot{Q}$ ratio of 1.0).[65] $\dot{V}_A/\dot{Q}$ inequality refers to regional mismatching of ventilation to perfusion. $\dot{V}_A/\dot{Q}$ inequality, no matter what its mechanism, interferes with overall efficiency of the lung for exchanging all gases, including oxygen, CO_2, CO, and anesthetic gases.[62]

$\dot{V}_A/\dot{Q}$ mismatch is the most common cause of hypoxemia. Several compensatory mechanisms tend to minimize the effects of abnormal $\dot{V}_A/\dot{Q}$ ratios. The low PA_{O_2} associated with a low $\dot{V}_A/\dot{Q}$ ratio causes pulmonary vasoconstriction. The low PA_{CO_2} associated with a high $\dot{V}_A/\dot{Q}$ ratio causes hypocapnic bronchoconstriction (e.g., pulmonary embolism).[66] These responses, however, only achieve partial compensation. As $\dot{V}_A/\dot{Q}$ inequality increases in the presence of a constant $\dot{V}_{O_2}$ and $\dot{V}_{CO_2}$, there is an immediate and marked fall in Pa_{O_2} and a slower increase in Pa_{CO_2}. The increase in Pa_{CO_2} and, to a lesser extent, the fall in Pa_{O_2} stimulates the chemoreceptors and leads to an increase in $\dot{V}_E$. In patients without a significant reduction in ventilatory capacity, the increase in ventilation is sufficient to bring Pa_{CO_2} back to normal, although it has only a small effect on the fall in Pa_{O_2} (Fig. 4-10). Thus, most patients with $\dot{V}_A/\dot{Q}$ inequalities have a low Pa_{O_2} but normal Pa_{CO_2}.[67] Ventilation in excess of normal alveolar requirement is termed *wasted ventilation.*[62] All normocapnic patients with COPD have increased ventilation of their alveoli, as do most hypercapnic patients.[62]

Figure 4-10



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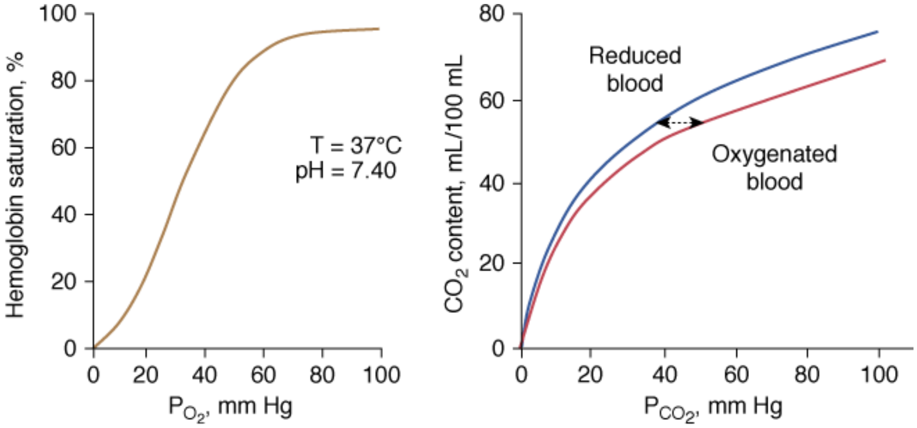
Effect of increasing overall ventilation Pa_{O_2} on and Pa_{CO_2} (lung model) as a function of different degrees of $\dot{V}_A/\dot{Q}$ mismatching, represented in terms of dispersion or standard deviations (σ) of the lognormal distribution of ventilation and perfusion. (Dispersion of 0.30 to 0.05 = normal $\dot{V}_A/\dot{Q}$ mismatch; 1.0 = moderate $\dot{V}_A/\dot{Q}$ mismatch; 2.0 = severe $\dot{V}_A/\dot{Q}$ mismatch.) Increases in overall ventilation have a powerful effect on Pa_{O_2} and Pa_{CO_2} when $\dot{V}_A/\dot{Q}$ dispersion is small. Abnormal $\dot{V}_A/\dot{Q}$ dispersion does not cause an increase in Pa_{CO_2} as long as patients are able to increase minute ventilation sufficiently. Pa_{O_2} also increases with increases in overall ventilation, although when $\dot{V}_A/\dot{Q}$ dispersions are (very) altered, normal Pa_{O_2} cannot be reached very easily, and further effects on ventilation have little effect on Pa_{O_2}. In the patients who cannot maintain a high rate of ventilation owing to the increased work of breathing and in those whose respiratory motor output increases only slightly when Pa_{CO_2} is high, hypercapnia can ensue. (Modified, with permission, from West.[64])

The different responses of Pa_{O_2} and Pa_{CO_2} to an increase in the level of ventilation is caused by the different shapes of the oxyhemoglobin and CO_2 dissociation curves (Fig. 4-11). The oxyhemoglobin dissociation curve is flat in the normal range. Thus, only units with moderately low $\dot{V}_A/\dot{Q}$ ratios benefit appreciably from the increased ventilation. Lung units that are positioned on the upper portion of the dissociation curve (high $\dot{V}_A/\dot{Q}$ ratio) develop little increase in the oxygen concentration of their effluent blood. The net result is that with increasing $\dot{V}_E$, the mixed Pa_{O_2} rises only modestly, and some hypoxemia always remains.[62] By contrast, the CO_2 dissociation curve is almost linear in the physiologic range (Fig. 4-11). Thus, an increase in $\dot{V}_E$ raises CO_2 output of lung units with both high and low $\dot{V}_A/\dot{Q}$ ratios.[62] The different shapes of the two dissociation curves are the main reason that patients with parenchymal lung disease have greater hypoxemia relative to hypercapnia. One final compensatory adjustment is possible: increase in cardiac output.[68] Adrenergic stimulation by arterial hypoxemia can raise cardiac output by 50% or more; this improves arterial blood gases by raising mixed venous oxygen and by lowering mixed venous CO_2.[68]

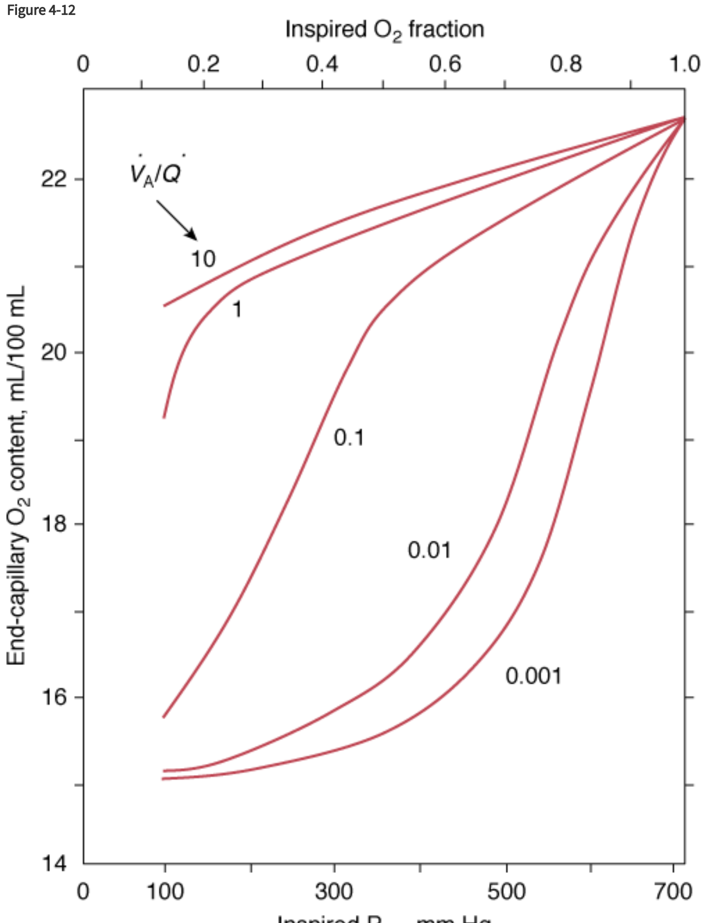
Figure 4-11



Left: Normal oxyhemoglobin dissociation curve. The curve has a sigmoid shape because when one subunit of the normal tetrameric form of the adult hemoglobin becomes oxygenated, it induces a structural change in the whole complex. Consequently, the three other subunits gain a greater affinity for oxygen until four molecules of oxygen are combined with hemoglobin. *Right:* The CO_2 dissociation curve for oxygenated and reduced blood. The relationship is steeper and more linear than the oxyhemoglobin dissociation curve. Oxygenation of blood causes the curve to shift to the right (Haldane effect), and, for a given CO_2 content, oxygenated blood has a higher P_{CO_2} than reduced blood.

Administration of supplemental oxygen in patients with $\dot{V}_A/\dot{Q}$ inequality will cause arterial hypoxemia to reverse impressively because $P_{A_{O_2}}$ of even poorly ventilated units increases sufficiently to achieve saturation (Fig. 4-12). Unless $F_{I_{O_2}}$ is 1.0, it is impossible to determine the relative contribution of right-to-left shunt versus $\dot{V}_A/\dot{Q}$ inequality to an increase in A-aD_{O_2}.[69] After breathing 100% oxygen for a sufficient time, only units that are totally or almost totally unventilated (shunt, true shunt, or anatomic shunt) will contribute to hypoxemia.[69]

Figure 4-12



The effect of alterations in inhaled partial pressure of oxygen (P_{O_2}) on oxygen content of end-capillary blood of a lung unit. Each line depicts a different ventilation–perfusion ratio ($\dot{V}_A/\dot{Q}$). With mild to moderate degrees of $\dot{V}_A/\dot{Q}$ inequality ($\dot{V}_A/\dot{Q}$ down to 0.1), end-capillary oxygen content increases as inhaled oxygen is increased. With severe $\dot{V}_A/\dot{Q}$ inequality ($\dot{V}_A/\dot{Q}$ below 0.1), the increase in end-capillary oxygen content with an increase in inhaled oxygen is much slower; only when inspired P_{O_2} is more than 400 to 500 mm Hg does end-capillary oxygen content of the lung unit reach values equivalent to those seen with mild to moderate degrees of $\dot{V}_A/\dot{Q}$ inequality. (Modified, with permission, from West.[64])

Systemic hypotension and hypertension modulate the ventilatory responses to hypoxemia (and hypercapnia). This so-called ventilatory baroreflex increases the operating point of the ventilatory response to hypoxemia (and hypercapnia) during hypotension and it decreases the operating point of the ventilatory response to hypoxemia (and hypercapnia) during hypertension.[70]

Physiologic Effects of Hypoxia

Although the physiologic effects of hypoxia are graded, the damaging effects are sudden.[71] A remarkable degree of arterial hypoxemia is required to cause tissue hypoxia.[25] In clinical practice, Campbell[25] observed that the lowest Pa_{O_2} compatible with life is 20 mm Hg (equivalent to an arterial oxygen saturation [Sa_{O_2}] of 30% to 40%). Evidence of end-organ damage is difficult to demonstrate in patients with a Pa_{O_2} above 40 mm Hg (equivalent to an Sa_{O_2} of approximately 70%).[25] Obviously, the duration of hypoxemia and the state of circulation (oxygen delivery) play major roles in determining the minimum Pa_{O_2} that does not cause end-organ damage or death.

The threshold Pa_{O_2} commonly used to diagnose hypoxemic respiratory failure is 60 mm Hg, which corresponds to an Sa_{O_2} of 90% (hypoxic hypoxia). Pa_{O_2} values below 60 mm Hg fall on the steep portion of the oxyhemoglobin dissociation curve, and decreases below that value are associated with precipitous falls in Sa_{O_2} (see Fig. 4-11). Although physiologically reasonable, for self-evident ethical reasons, the 60 mm Hg (Pa_{O_2}) threshold cannot be validated experimentally.

The main concern with hypoxemia is impaired tissue oxygenation, especially of the heart and brain. The factors determining oxygen supply to the tissues include hemoglobin concentration, Sa_{O_2}, the affinity of hemoglobin for oxygen (P_{50}), cardiac output, regional oxygen consumption-to-perfusion relationships, and the diffusion of oxygen from the capillary to intracellular sites. The amount of oxygen delivered to the tissues is calculated as

$$O_2 \text{ delivery} = Ca_{O_2} \times \text{cardiac output}$$

where arterial oxygen content (Ca_{O_2}) is calculated as

$$Ca_{O_2} = (Hb \times 1.34 \times Sa_{O_2}/100) + (0.003 \times Pa_{O_2})$$

Even with a satisfactory Pa_{O_2}, tissue hypoxia may arise because of decreased Ca_{O_2} (e.g., decreased hemoglobin concentration or decreased hemoglobin function, such as in carbon monoxide poisoning or anemic hypoxia), decreased oxygen delivery (e.g., cardiogenic shock or stagnant hypoxia), and decreased capacity of the tissues to use oxygen (e.g., sepsis, cyanide intoxication, or histotoxic hypoxia). Otherwise stated, tissue hypoxia can be present despite adequate Pa_{O_2}, or it can be absent despite an abnormally low Pa_{O_2}.[59]

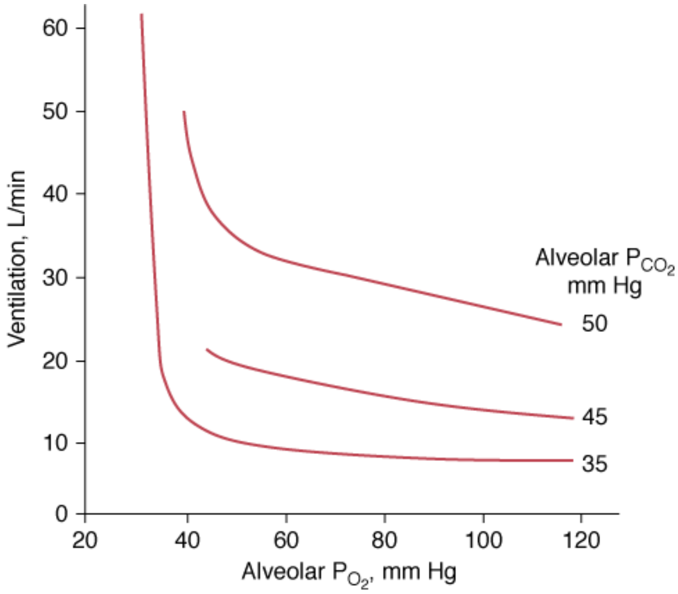
Respiratory Responses

Peripheral chemoreceptors (carotid and aortic bodies) detect changes in arterial oxygen. Within seconds after the onset of hypoxia, they initiate reflexes that are important for maintaining homeostasis.[72–74] The aortic bodies play a minor role in modulating spontaneous respiratory activity, although they have a discernible effect when their gain is increased by hypercapnia.[75]

Hypoxia augments sensory discharge from the peripheral chemoreceptors, which, in turn, send neural impulses to the respiratory centers (inspiratory neurons of the dorsal respiratory group and ventral respiratory group[76]), causing an increase in the $\dot{V}_E$.[72,73,77,78] Hyperpnea, in turn, activates pulmonary afferents, thereby buffering the sympathetic response to hypoxemia.[79]

The ventilatory response to progressive hypoxia is hyperbolic[72] and increases in the presence of concurrent hypercapnia[75] (Fig. 4-13). It decreases with age.[80] Chronic hypoxia may induce a reduced ventilatory response to hypoxia in patients with COPD,[75] although the role of airway narrowing in producing this effect cannot be excluded.[75]

Figure 4-13



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Relationship between ventilation and alveolar pressure of oxygen (P_{O_2}) at various levels of alveolar carbon dioxide (P_{CO_2}). The ventilatory response increases with the increase in P_{CO_2}. At alveolar P_{CO_2} 35 mm Hg, ventilation exhibits virtually no increase until alveolar P_{O_2} falls below 40 mm Hg and, thereafter, ventilation increases abruptly.

The ventilatory response to hypoxemia is attenuated or abolished in patients who have undergone surgical excision of the carotid bodies.[73,74,78,81] The increase in ventilation in response to hypoxemia probably contributes to coronary vasodilation.[82] Stimulation of the carotid bodies with nicotine under normoxic conditions causes an increase in ventilation and coronary vasodilation.[82] Coronary vasodilation does not occur if the increase in ventilation is prevented by general anesthesia.[82]

Cardiovascular Responses

Hypoxic stimulation of chemoreceptors triggers reflex adrenergic vasoconstriction in muscle and coronary vasodilation but does not elicit a reflex response in the cerebral vessels.[74,79,82,83] Hypoxia also causes local vasodilation.[72,79] The net effects are increases in heart rate, cardiac output (resulting from the positive chronotropic effect of hypoxemia, not increased stroke volume[84]), pulmonary artery resistance,[85] and cerebral and coronary blood flow.[74,79,80,82–84] Hypoxemia fails to increase systemic blood pressure[84] or increases it very modestly (<10 mm Hg rise in systolic and diastolic pressures).[77] The increase in blood pressure, but not in heart rate, is absent in patients in whom the carotid bodies are inactivated.[74,77] Persistent tachycardic response to hypoxemia in patients with bilateral carotid body ablation likely is mediated by the effect of hypoxemia on the aortic bodies.[72]

Tachycardia caused by hypoxemia is mediated by multiple factors, including CNS-mediated sympathetic discharge, effect of P_{O_2} on the cardiac pacemaker, and concomitant hyperpnea.[77] These mechanisms presumably override the cardioinhibitory signals (bradycardic effect) from the carotid body.[77] The importance of hyperpnea in overriding the bradycardic effect is supported by the observation that hypoxemia combined with cessation of lung inflation (sleep apnea,[86] breath holding,[87] or neuromuscular blockade during intubation[88]) or diminution of lung inflation (such as when hyperpnea is prevented by controlled ventilation[89,90]) is more apt to produce bradycardia than tachycardia.[72,86] Bradycardia results from the concurrent activation of the carotid bodies by hypoxia and increased cardiac vagal activity induced by apnea.[79,83] The increased vagal activity (induced by apnea) can be so intense as to induce bradycardia in the absence of coexisting hypoxemia; this response has been described in patients with spinal cord injury within seconds of discontinuation from a ventilator.[91] Simultaneously with the increase in cardiac vagal activity, diminution of lung inflation causes marked potentiation of the sympathetic vasoconstrictor response to hypoxemia (secondary to the lack of inhibitory influence of the pulmonary stretch receptors).[83] (This combination of sympathetic vasoconstriction and vagal bradycardia constitutes part of the diving reflex.[83])

Severe hypoxemia is not tolerated by the CNS because it has a high rate of oxygen consumption and lacks alternative energy reserves.[92] Therefore, severe hypoxemia causes cerebral depression.[73] Patients with cerebral hypoxia develop bradycardia, hypotension,[88] and hypoventilation,[93] which further worsens hypoxia and induces a potentially lethal vicious cycle.

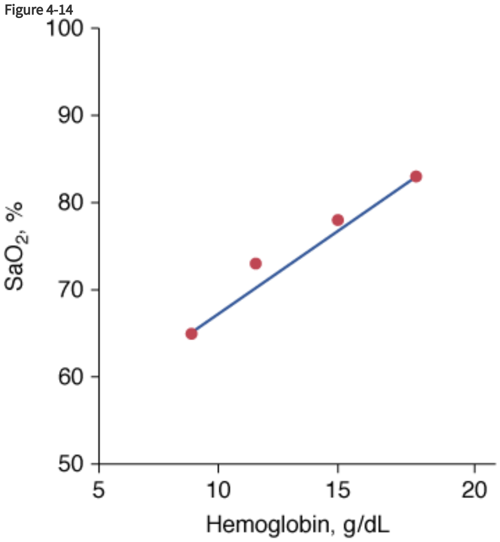
Clinical Presentation of Hypoxemia

Cyanosis

Physicians commonly view cyanosis as the hallmark of hypoxemia.[72] Cyanosis is recognized as a blueness of the capillary blood visible through the mucous membranes or skin, where capillaries are numerous and close together and the tissues over them are thin and transparent, such as the lips (central cyanosis) and nail beds.[72]

Based on the classic work of Lundsgaard and Van Slyke, it is widely believed that central cyanosis represents the presence of at least 5 g/dL of deoxygenated hemoglobin in the blood of the capillaries.[94] Because the arteries and most of the veins are so far away from the skin, their content cannot influence skin color.[94] Importantly, it is the quantity of reduced hemoglobin per deciliter of capillary blood, not the relative lack oxygenated hemoglobin, that produces the blue coloration of cyanosis.[95] Lundsgaard and Van Slyke demonstrated the crucial effect of hemoglobin on the development of cyanosis.[94] When hemoglobin is 15 g/dL, cyanosis reflects an Sa_{O_2} of 78%. In an anemic patient with a hemoglobin of 9 g/dL, cyanosis will not occur until Sa_{O_2} falls to 65%, whereas in a polycythemic patient, with a hemoglobin of 18 g/dL, cyanosis will occur at a Sa_{O_2} of 83% (Fig. 4-14). In addition to hemoglobin, detection of cyanosis also depends on variables such as thickness and opacity of the skin and perfusion status, as well as on the visual skills of the observer.[96,97] In seventy-two patients who had Sa_{O_2} values ranging between less than 75% and 100%, agreement among three observers was only 69% when inspecting the tongue, 61% when inspecting the lips, and 53% when inspecting the nail bed.[97] In that study,[97] cyanosis of the tongue and lips was recorded in 40% to 45% of patients with normal Sa_{O_2} values. Rather than a blue or pink color, pallor is seen with both anemia and shock. Thus, severe or fatal tissue hypoxia may occur without cyanosis.[72]

Figure 4-14



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

The dependence of cyanosis—resulting from the presence of 5 g of reduced hemoglobin per deciliter of capillary blood—on prevailing hemoglobin. In anemic patients, cyanosis signifies more severe hypoxemia than in a polycythemic patient. (Based on data from Lundsgaard and Van Slyke.[94])

Cardiopulmonary Manifestations

Acute hypoxemia increases respiratory frequency, tidal volume, and $\dot{V}_E$ in almost all subjects.[72] Increases in $\dot{V}_E$, however, are an unreliable guide because the interindividual response is considerable.[96] The standard deviation for the within-subject variability in day-to-day hypoxic ventilatory response is approximately 22%.[98] The respiratory response to mild or moderate hypoxemia may be barely measurable or absent.[72] An increase in heart rate is equally of limited value because many other factors, such as fever, low blood pressure, pain, apprehension, and drugs, can cause it to rise.[72] Moreover, under specific circumstances (see "Cardiovascular Responses" above), hypoxemia causes bradycardia rather than tachycardia. Hypoxemia tends to increase systolic blood pressure in some subjects; the wide variations in response[77] renders arterial pressure of little value in the diagnosis of hypoxemia.[72]

Neurologic Manifestations

The metabolic needs of the brain largely depend on oxidation of glucose to CO_2 and water.[92] The brain cannot store oxygen. It survives only for minutes after the supply is reduced to critical levels.[92] In acute anoxia, consciousness is lost within 15 seconds. The electroencephalogram slows with a Pa_{O_2} of less than 35 mm Hg or with blood flow of less than 40% of normal.[92] Loss of the electroencephalogram tracing occurs when cerebral P_{O_2} reaches 20 mm Hg or following 20 seconds of complete anoxia.[92]

In the absence of defects of cerebral blood flow, Pa_{O_2} below 40 mm Hg is required to produce prominent symptoms (Table 4-3).[24,99,100] Confusion and delirium (resembling alcohol intoxication) appear at Pa_{O_2} values of 35 to 50 mm Hg.[65] Tremors and asterixis (flapping tremor elicited by dorsiflexing the wrists with the arms outstretched and caused by a momentary interruption of normal continuous action potentials to both flexor and extensor muscles) are infrequent even when the Pa_{O_2} is less than 40 mm Hg.[24,25] Somnolence and obtundation occur at Pa_{O_2} values of 25 to 35 mm Hg. Some patients develop myoclonic jerks (bursts of excitation to resting muscles) and seizures.[24] At approximately 25 mm Hg, consciousness is lost,[65] and death often ensues.

Table 4-3: Neurologic Signs and Symptoms of Hypoxia

Pa_{O_2}, mm Hg	Signs and Symptoms of Hypoxia
35 to 50	Loss of critical judgment, confusion, delirium (resembling alcohol intoxication), tremors, asterixis
25 to 35	Somnolence, obtundation, myoclonic jerks, seizures
20 to 25	Loss of consciousness
<20	Death

When considering the neurologic manifestations of hypoxemia, Pa_{O_2} is only a small part of a complex situation. A decrease in Pa_{O_2} may be well tolerated if oxygen delivery is maintained by the combination of increased cardiac output and systemic vasoconstriction.[101] These compensatory mechanisms, however, can be overwhelmed by anemia or carbon monoxide poisoning, which decrease oxygen-carrying capacity (anemic hypoxia), or by atherosclerosis or other causes of vascular occlusion (ischemic hypoxia) in which the increased cardiac output does not suffice to prevent tissue damage.

Hypercapnic Respiratory Failure

Pathophysiology

Hypercapnic respiratory failure is a state in which ventilation is insufficient to maintain a normal Pa_{CO_2} for the level of metabolic activity (measured by CO_2 production, $\dot{V}_{CO_2}$).[1] Common causes include COPD, severe asthma, conditions where respiratory motor output is decreased (e.g., neoplasm and infections of the CNS, medications, and drugs), neuromuscular-skeletal diseases (e.g., myasthenia gravis, Guillain-Barré syndrome, and trauma), and upper airway obstruction.

Under steady-state conditions, the relationship between Pa_{CO_2}, alveolar ventilation ($\dot{V}_A$), and $\dot{V}_{CO_2}$ is given by the equation

$$Pa_{CO_2} = (\dot{V}_{CO_2}/\dot{V}_A) \cdot K$$

The constant K is usually stated as 0.863; it converts measurements of $\dot{V}_{CO_2}$ from standard conditions to body-temperature conditions. The term $\dot{V}_A$ represents the portion of $\dot{V}_E$ that reaches the terminal gas-exchange units and is calculated as

$$\dot{V}_A = \dot{V}_E - V_D$$

where V_D equals dead space ventilation. A reduction in $\dot{V}_A$ may result from an inadequate $\dot{V}_E$ or an increase in V_D (resulting from an increase in true V_D or a functional increase in V_D secondary to lung regions with high $\dot{V}_A/\dot{Q}$ relationships).

The mechanisms responsible for hypercapnia can be grouped into two categories, depending on whether there is (or is not) an increased $A\text{-}aD_{O_2}$. In pure alveolar hypoventilation (e.g., neuromuscular diseases, drug overdoses, and CNS pathologies), $A\text{-}aD_{O_2}$ is usually normal (unless lung abnormalities are present). In disorders associated with $\dot{V}_A/\dot{Q}$ inequality (e.g., COPD and ARDS), $A\text{-}aD_{O_2}$ is increased.

Pa_{CO_2} in excess of 90 mm Hg is unlikely in patients breathing room air because the concomitant degree of hypoxia is incompatible with survival.[25] Such Pa_{CO_2} values can occur if a patient is breathing oxygen-enriched air.[102]

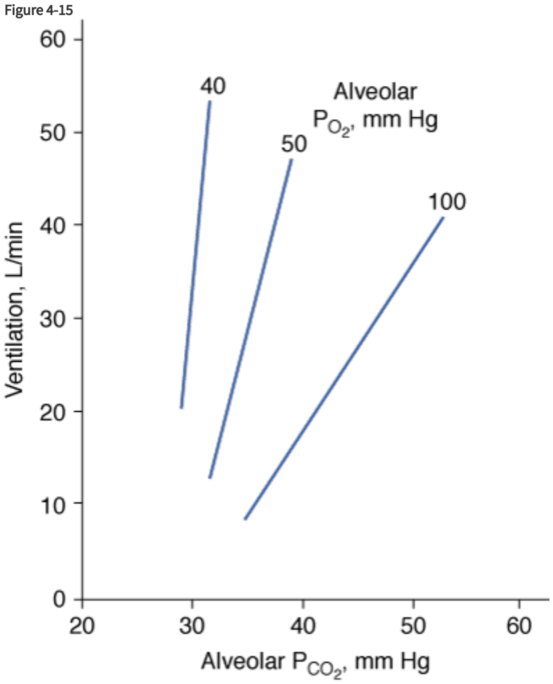
Physiologic Effects of Hypercapnia

Hypercapnia elicits autonomic and ventilatory responses primarily through central chemoreceptors located in the rostral ventrolateral medulla.[74] These respond to changes in hydrogen ion concentration.[74] Hypercapnia, probably via a reduction in intracellular pH, also stimulates peripheral arterial chemoreceptors located in the carotid bodies, aortic bodies, and abdomen.[103,104] Only 15% to 30% of the ventilatory response to hypercapnia results from peripheral chemoreceptor stimulation.[105,106] Not surprisingly, the Pa_{CO_2} of patients with bilateral resection of the carotid bodies is higher (by 4.6 ± 1.3 [SD] mm Hg) than in healthy subjects.[81]

Respiratory Responses

Hypercapnia causes an increase in $\dot{V}_E$; stimulation peaks with an inhaled CO_2 of 10%.[107] In contrast to the hyperbolic response to progressive hypoxemia (see Fig. 4-13), the hypercapnic ventilatory response is linear (Fig. 4-15).[72] The ventilatory response to CO_2 is enhanced in the presence of hypoxia or metabolic acidosis[108] and decreases with age.[80]

Figure 4-15



Ventilatory response to progressive hypercapnia. Ventilation increases linearly with increase in alveolar carbon dioxide (P_{CO_2}). Decreases in alveolar oxygen (P_{O_2}) produce a steeper ventilatory response to progressive hypercapnia.

Hypercapnia and hypoxia induce different patterns of neuromuscular activation as $\dot{V}_E$ rises—even when the respiratory components of tidal breathing (tidal volume and inspiratory and expiratory times) are similar.[76] First, hypercapnia is a more potent stimulus for expiratory muscle recruitment than is hypoxemia—one-third of subjects do not recruit their expiratory muscles during hypoxemia.[76] Second, activation of the diaphragm is greater during hypoxemia than during hypercapnia.[76] (The lack of expiratory recruitment during hypoxemia may increase end-expiratory lung volume, which increases oxygen reserves.[76]) The ventilatory response to CO_2 in healthy subjects exhibits a very wide range, 0.47 to 8.16 L/min/mm Hg, although approximately 80% of subjects have a response between 1.5 and 5 L/min/mm Hg.[1]

Cardiovascular Responses

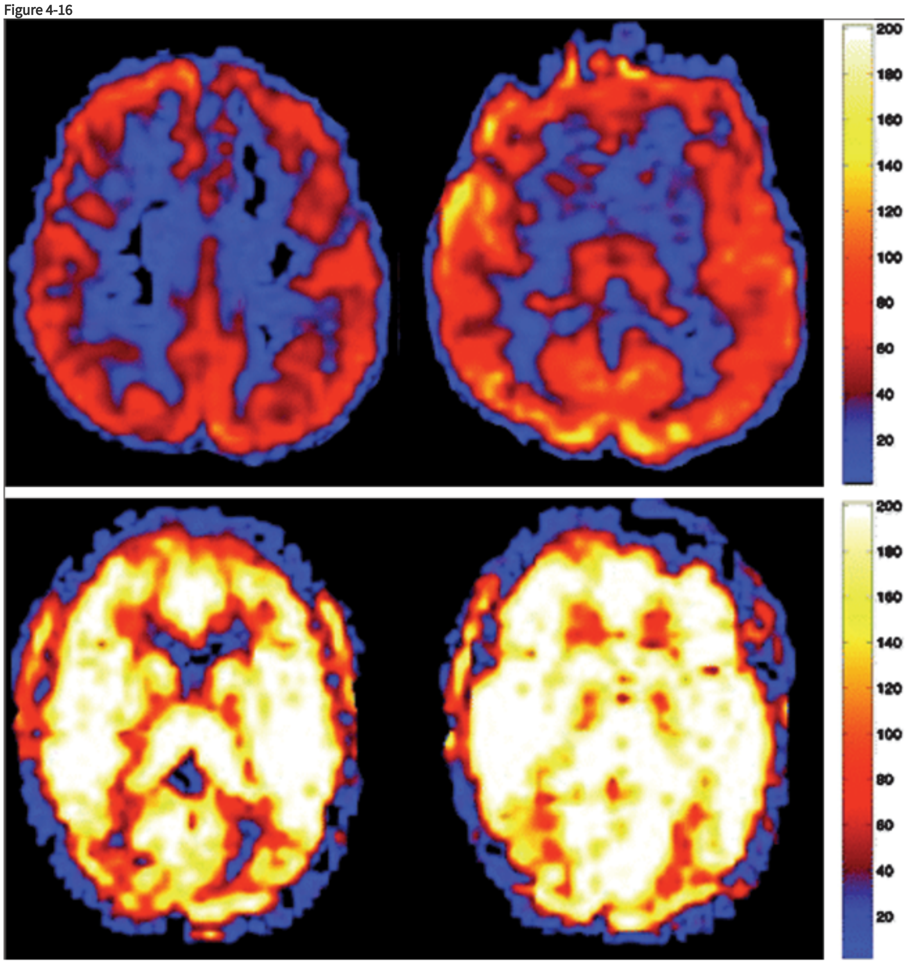
Hypercapnia causes greater increases in sympathetic activity[104,109] and (usually) greater increases in systemic blood pressure (approximately 30 mm Hg rise in systolic pressure and approximately 25 mm Hg rise in diastolic pressure[110]) than does hypoxemia.[104] Acting via baroreflexes, this greater hypertensive response may be partly responsible for the more limited rise in heart rate during hypercapnia than during hypoxemia.[104] The tachycardic response to hypercapnia is blunted in the elderly.[80] Apnea increases the sympathetic nerve activity elicited by hypercapnia.[104] This increase, however, is less than the increase during hypoxemia.[104]

Combined hypoxia and hypercapnia have a synergistic effect on sympathetic nerve activity[104] and hyperpnea. This potentiation may arise because hypercapnia sensitizes the response of peripheral chemoreceptor afferents to hypoxia.[104] Another possibility is that both peripheral and central chemoreceptors synapse on common nuclei in the brainstem.[104]

Hypercapnia not only has a sympathetic vasoconstrictor effect (secondary to chemoreceptor activation) but also has a direct vasodilator effect on systemic arterioles[23,72]; dilation of conjunctival and superficial facial vessels may be noted. The first action, however, is predominant in conscious persons, and blood pressure and heart rate increase. If the vasomotor center cannot respond (e.g., secondary to brain damage, severe ischemia or hypoxia, or deep anesthesia) or is disconnected from peripheral parts of the sympathetic nervous system (e.g., secondary to spinal cord damage or blocking drug or spinal anesthesia), direct vasodilation becomes the sole or dominant effect, and blood pressure falls.[23,72,111]

Hypercapnia causes cerebral hyperperfusion (Fig. 4-16).[112] The increase in blood flow is proportional to the severity of hypercapnia.[112] Cerebrovascular reactivity to CO_2 depends on age (i.e., there is reduced cerebral perfusion reserve in the elderly)[113] and state (i.e., there is a 70% reduction in cerebrovascular reactivity to CO_2 during non–rapid eye movement sleep).[114] Hypercapnic cerebrovascular reactivity also is reduced in patients with preexisting cerebrovascular diseases.[115]

Figure 4-16



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Arterial spin-labeled magnetic resonance perfusion imaging. *Upper panel:* A patient with normocapnia (Pa_{CO_2} 39 mm Hg), in whom the left image passes through the corona radiata and the right through the thalamus. Cerebral blood flow throughout the gray matter (red discoloration) is normal (mean: 63 mL/100 g tissue/min). *Lower panel:* A patient with emphysema and hypercapnia (Pa_{CO_2} 56 mm Hg), in whom the left image passes through the thalamus and the right through the basal ganglia. Cerebral blood flow is markedly increased (white-yellow discoloration; mean: 175 mL/100 g tissue/min), signifying global hyperperfusion. (Modified, with permission, from Pollock et al.[112])

Neurologic Responses

Hypercapnia decreases cerebral metabolic rate for glucose and interferes with cerebral energy production.[116] The cerebral metabolic rate for oxygen is maintained, or slightly increased, provided that Pa_{CO_2} is less than 90 to 100 mm Hg.[116] For higher values of Pa_{CO_2}, cerebral metabolic rate for oxygen decreases.[116] Hypercapnia has a dual effect on neuron excitability: stimulatory at low concentrations and inhibitory at high concentrations.[72] In humans, very high concentrations (30%) can produce surgical anesthesia, which can be associated with seizures.[72,102,107]

Clinical Presentation of Hypercapnia

The clinical manifestations of hypercapnia result from a complex interaction of several factors, including severity of hypercapnia, comorbidities, and the speed at which the increase in CO_2 has occurred. For example, patients receiving chronic oxygen therapy have been reported to function satisfactorily with Pa_{CO_2} values of greater than 100 mm Hg.[72] Therefore, there is no single threshold of Pa_{CO_2} above which mechanical ventilation is mandatory.

Most signs and symptoms of acute hypercapnic respiratory failure, including hyperpnea, dyspnea, tachycardia and hypertension, and diaphoresis, are similar to those of hypoxemia. Some consider it a waste of effort to try to separate which manifestations are related to hypoxemia and which to hypercapnia.[117]

The major clinical features of hypercapnia are those affecting the CNS. One difference between acute hypoxemic and acute hypercapnic respiratory failure is the greater incidence of neurologic manifestations with the latter. Acute hypercapnia can cause fine tremors (of the outstretched hands, head, or legs), asterixis, myoclonic jerks, sustained myoclonus, and seizures.[24,107] It also can cause cognitive disorders, hostility, irritability, paranoid behavior, somnolence, stupor, and coma.[24] In a study of thirty-two episodes of acute respiratory failure, Kilburn[24] reported that the severity of cognitive disorders, asterixis, and somnolence, and the presence of stupor and coma were closely related to the severity of respiratory acidosis—and not to the severity of hypoxemia.

Some patients with severe hypercapnia have papilledema and elevated cerebrospinal fluid pressure probably because of the increase in blood volume within the near-rigid cranial cavity.[72,118] Under conditions of prolonged hypercapnia (several hours), cortical blood flow may return toward baseline over time.[119] The latter is probably mediated by a buildup of brain extracellular bicarbonate and an increase in pH.[119]

Some patients with combined hypoxemic and hypercapnic respiratory failure become more comatose when treated with oxygen (CO_2 narcosis). The mechanisms responsible for oxygen-induced hypercapnia are complex and probably include reduction in ventilation, increased wasted ventilation (alveolar dead space),[120] and the Haldane effect (Pa_{CO_2} increases because of net release of CO_2 from erythrocytes when Sa_{O_2} is increased)[1] (see Fig. 4-11).

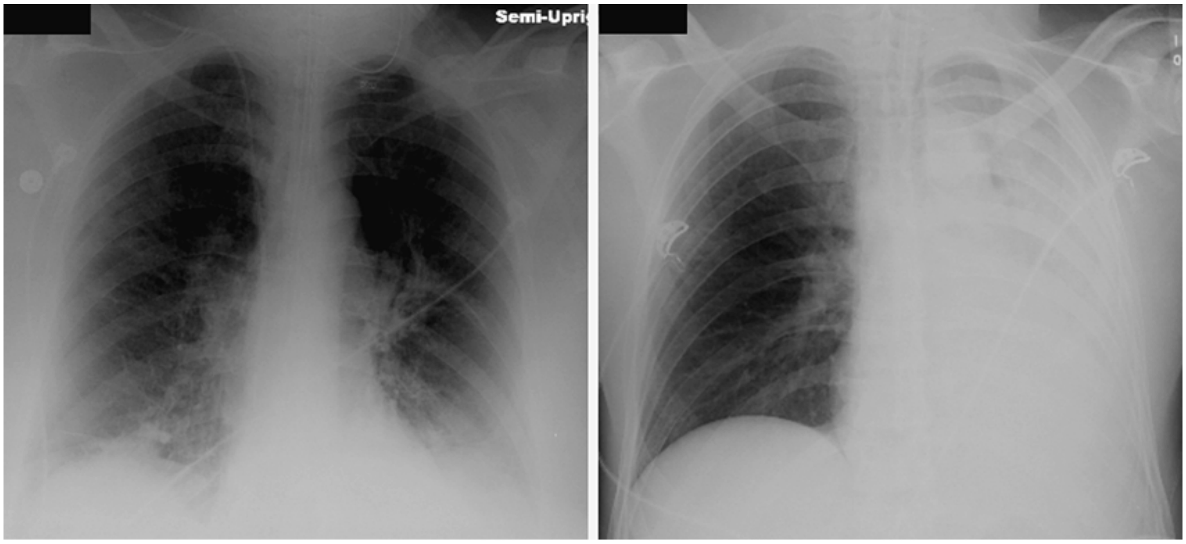
Postoperative Respiratory Failure

Postoperative respiratory failure can be defined as the need for intubation and mechanical ventilation in the 48 hours after surgery.[121] Among more than 180,000 patients who underwent major noncardiac surgery, postoperative respiratory failure occurred in 3%.[121] This exceeds the incidence of cardiac arrest or acute myocardial infarction after noncardiac surgery (1.3%).[122] Among 1055 patients, most of whom underwent lower abdominal/inguinal hernia repair or orthopedic limb surgery, 0.1% required intubation within 7 days of surgery.[123]

Pulmonary complications are estimated to account for nearly 25% of deaths within 6 days of surgery.[124] This figure may be an underestimate; many patients with respiratory failure can be kept alive by ventilator support, only to die from nonrespiratory complications (e.g., sepsis and multiorgan failure).[125] Among patients who undergo major noncardiac surgery, mortality is 27% to 42% for those with postoperative respiratory failure versus 1% to 6% for those without postoperative respiratory failure.[121,126,127] Respiratory failure is the most important determinant of postoperative mortality in 40% to 100% of thoracic surgery patients.[125,128,129]

A common cause of postoperative respiratory failure is the development of atelectasis (Fig. 4-17). Atelectasis is the most frequent pulmonary complication after general surgery (particularly thoracic and upper abdominal surgery) (Table 4-4).[125,128,129] Atelectasis occurs in approximately 90% of patients during anesthesia.[130] During uneventful anesthesia, before any surgery is begun, 15% to 20% of the lung base is collapsed.[131] With thoracic surgery and cardiopulmonary bypass, more than 50% of the lung can be collapsed several hours after surgery.[132] After abdominal surgery, atelectasis can persist for several days,[133] and application of lung recruitment maneuvers and PEEP at the conclusion of the procedure do not improve postoperative oxygenation.[134]

Figure 4-17



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Postoperative atelectasis. *Left:* Portable anteroposterior chest radiograph showing bibasilar subsegmental atelectasis following hemicolectomy. *Right:* Atelectasis of the entire right lung caused by mucus plugging following drainage of submandibular and pterygomandibular space abscesses.

Table 4-4: Causes of Postoperative Respiratory Failure

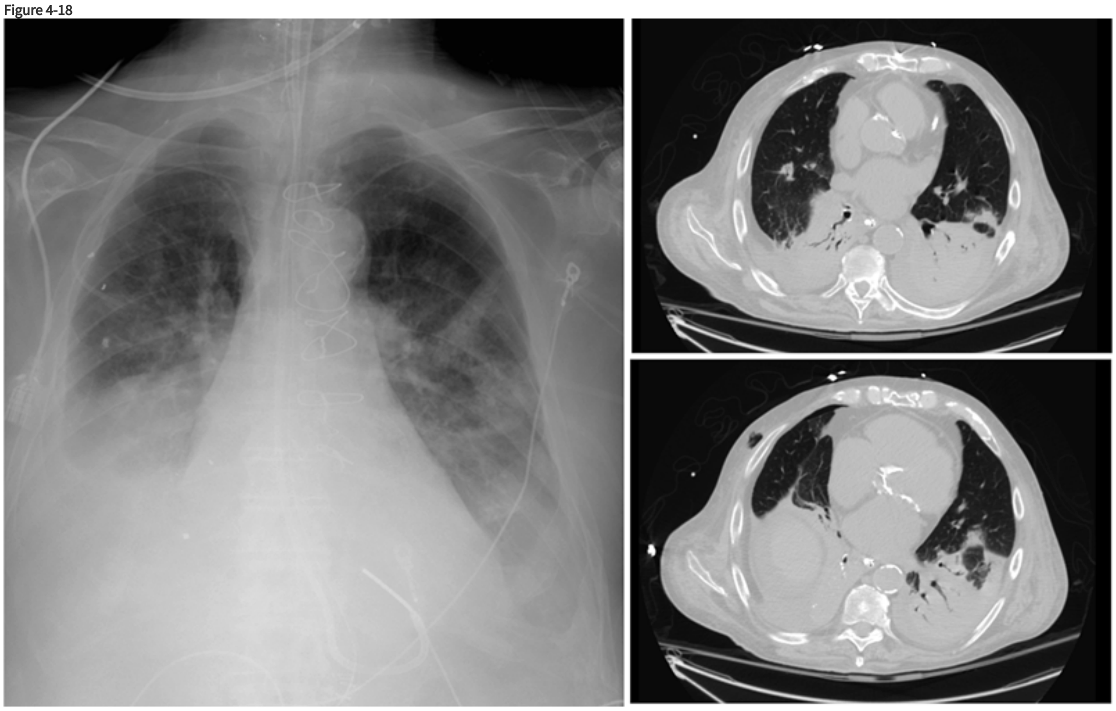
Intrapulmonary causes
Atelectasis
Aspiration
Pneumonia
Acute respiratory distress syndrome/acute lung injury
Volume overload/congestive heart failure
Pulmonary embolism (thrombus, air, fat)
Bronchoconstriction (asthma/COPD)
Pneumothorax
Extrapulmonary causes
Shock
Sepsis
Decreased respiratory motor output
Phrenic nerve injury
Diaphragmatic dysfunction
Upper airway obstruction
Obstructive sleep apnea

Abbreviation: COPD, chronic obstructive pulmonary disease.

One mechanism for the development of atelectasis during general anesthesia is decreased respiratory muscle tone, which is accompanied by cephalad displacement of the diaphragm, 20% reduction in functional residual capacity (60% reduction in obesity), and compression of lung tissue.[125,135] Two other purported mechanisms are impaired function of surfactant and resorption of gas behind occluded airways (particularly with high $F_{I_{O_2}}$).[131]

Persistent or new atelectasis after anesthesia can be caused by several mechanisms. First, patients undergoing abdominal or thoracic surgery experience a marked reduction in vital capacity and a smaller but clinically important decrease in functional residual capacity.[135] These changes are ascribed to postoperative pain and diaphragmatic dysfunction.[135,136] Abnormal abdominal mechanics reduce end-expiratory lung volume below (the increased) closing volume, leading to absorption of gas from poorly ventilated lung units.[137] Second, the weakened diaphragm no longer acts as a rigid wall between the thoracic and abdominal space.[136] The positive abdominal pressure transmitted to the thoracic cavity increases pleural pressure and causes compression atelectasis—particularly in the dependent thorax.[131,138] Third, mucociliary clearance is impaired.[139,140] Fourth, narcotics[141] and pain[142] suppress periodic deep breaths. Lack of intermittent deep breaths favors alveolar collapse by precluding alveolar recruitment and decreasing active forms of alveolar surfactant.[143] Fifth, narcotics[141] and pain[142] also suppress cough and interfere with the ability to clear secretions.[144] In a prospective study of 361 patients undergoing elective lung surgery (including pneumonectomy, lobectomy, wedge and segmental resection, and bullectomy), Bonde et al[144] reported that complications related to retention of secretions (i.e., atelectasis, pneumonia, and respiratory failure) occurred in 30% of patients.[144] On multivariate analysis, being a current smoker, having ischemic heart disease, and the absence of regional analgesia or failure of regional analgesia (thoracic epidural or extrapleural intercostal nerve infusion block) increased the risk of secretion retention.[144] Of twelve in-hospital deaths, ten were considered complications of secretion retention. Sixth, risk of atelectasis is increased with routine use of nasogastric tubes.[123,145,146] Seventh, preexisting pulmonary disease[147] and current smoking[148]—associated with increased postoperative secretions[144]—increase the risk of atelectasis. The end result of atelectasis can be hypoxemic respiratory failure, hypercapnic respiratory failure, or both, and pneumonia and sepsis (Fig. 4-18).[137,149] Development of mucous plugging with lobar or multilobar collapse of the lung is a less-common cause of atelectasis than the development of subsegmental atelectasis (see Fig. 4-17).

Figure 4-18



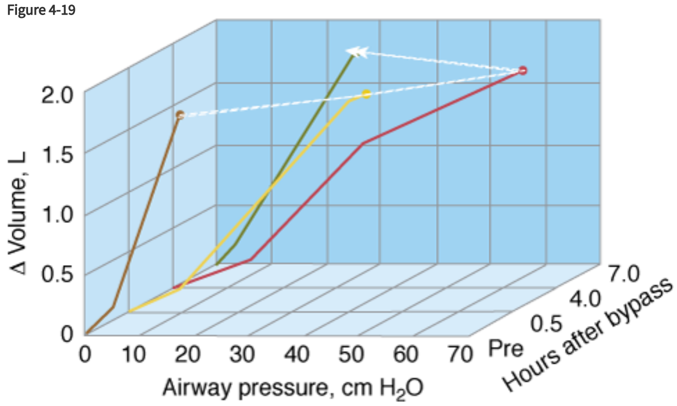
Intrapulmonary and *extrapulmonary* causes of postoperative respiratory failure. Eighty-year-old man with COPD and aortic valve replacement underwent small bowel resection and repair of traumatic diaphragmatic hernia. Six days after surgery he developed respiratory distress caused by atelectasis, atrial fibrillation, volume overload, and aspiration. Following intubation, chest radiograph *(left)* demonstrated discoid atelectasis in the left upper lung zone, pleural effusions, and pulmonary infiltrates. Computed tomography *(right)* revealed air space opacities suggestive of pneumonia and lower-lobe atelectasis.

Preoperative assessment is useful in identifying patients at increased risk after thoracic and nonthoracic surgery.[123,126,150] In a prospective study of 272 patients undergoing nonthoracic surgery, McAlister et al[150] identified three independent risk factors for postoperative complications: age 65 years or older, smoking of 40 pack-years or more, and maximum laryngeal height of 4 cm or less.

In a subsequent study of 1055 patients undergoing nonthoracic surgery, these investigators reported four independent risk factors for postoperative complications: age 65 years or older, positive cough test (recurrent coughing after asking a patient to cough once), perioperative nasogastric tube, and duration of anesthesia of 2.5 hours or longer.[123] In a prospective study of more than 180,000 patients undergoing major noncardiac surgery, Arozullah et al[126] reported that type of surgery, albumin and blood urea nitrogen levels, functional status, COPD, and age could be used to generate an index to identify patients at risk for postoperative respiratory failure. These results were prospectively validated by the same investigators in a second study comprising more than 180,000 patients undergoing major noncardiac surgery.[121] Solid data on the usefulness of respiratory physiotherapy, including incentive spirometry, inspiratory muscle training, and noninvasive ventilation to *prevent* pulmonary complications, after cardiac or upper abdominal surgery are lacking.[151]

Patients undergoing cardiac surgery are especially prone to postoperative complications because they are subjected to prolonged anesthesia and, commonly, hypothermia.[152,153] Patients also often require therapy for hypotension or hypertension, as well as fluid resuscitation (including blood transfusions).[152,153] Cardiac surgery can temporarily increase respiratory load.[154–156] At the end of coronary bypass surgery, lung compliance is less and lung resistance greater after chest closure than before surgery.[154] An increase in lung water after cardiopulmonary bypass, especially if the lungs remain collapsed during surgery, contributes to the worsening mechanics. In eight patients undergoing valvular surgery, compliances of the chest wall and lung were less at 4 hours after surgery than before surgery (Fig. 4-19).[155] By 7 hours, chest wall compliance was back to baseline, and lung compliance was higher than before surgery.[155] The investigators speculated that the initial decrease in lung compliance is caused by interstitial fluid secondary to increased vascular permeability.[155] The subsequent increase in lung compliance may result from mobilization of fluid that had accumulated before surgery (as a consequence of valvular disease) and as a result of extracorporeal circulation (increased permeability). Severe restrictive pulmonary defect is the rule,[157,158] and venous admixture (or sum of true shunt and $\dot{V}_A/\dot{Q}$ mismatch) is increased.[158] At the end of surgery, many patients are transferred to the ICU while fully ventilated. Patients then are given the time to rewarm, to metabolize the medications received during anesthesia, and to receive therapy for any hemodynamic derangements that are present.

Figure 4-19



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Cardiopulmonary bypass causes a reversible impairment in respiratory mechanics. Static inflation volume-pressure (V-P) curve of the respiratory system in a patient before cardiopulmonary bypass (*Pre*) and 0.5, 4.0, and 7.0 hours later. Compared with before bypass (*orange curve*), the slope of the V-P curve is less steep 0.5 hours later (*yellow curve*), and less steep again at 4 hours (*red curve*). At 7 hours, the slope of the V-P curve has returned to baseline (*green curve*). An increase in lung water after bypass contributed to the temporary worsening of compliance. Δ *Volume,* changes in lung volume relative to the elastic equilibrium volume of the respiratory system. (Data from Ranieri et al.[155])

From the 1960s to 1990s, prolonged controlled mechanical ventilation was the standard of care following cardiac surgery.[159–161] This strategy was justified by the use of high-dose narcotic anesthesia and concerns about myocardial ischemia in the early postoperative period.[161–163] Since the early 1990s, under pressure of cost containment and improved resource utilization,[164] early extubation strategies have been implemented successfully in uncomplicated cardiac surgery cases.[152,153,165] Early extubation is achieved by modifying intraoperative anesthetic techniques (e.g., usually a decrease in total opioid administration, use of ultrashort-acting opiates, and use of inhaled techniques[153,166,167]), minimizing sedation during the postoperative ICU stay,[153] and improving postoperative pain control.[168] Recovery to spontaneous respiration and extubation is shorter with rapid tracheal extubation strategies than with conventional strategies—median time of approximately 4 and 7 hours, respectively.[152,153] Approximately 20% to 30% of patients, however, do not tolerate (or are not candidates for) rapid tracheal extubation strategies because of postoperative complications (e.g., bleeding, myocardial ischemia, myocardial infarction, refractory hypoxemia, or neurologic complications).[152,153] Early extubation may improve cardiac output[169] and renal perfusion[162,170] and reduce cardiopulmonary morbidity,[171] as well as decrease hospital stay without adverse outcome.[165,166]

In addition to early extubation (extubation achieved in less than 6 hours after surgery[165]), immediate extubation in the operating room has been reported after coronary bypass grafting with or without valve replacement,[161,167,172,173] single-lung transplantation,[174] and two-stage esophagectomy.[175] Recently, coronary bypass grafting with high thoracic epidural anesthesia in the awake and spontaneously breathing patient has been achieved successfully.[176] This latter strategy remains highly controversial.[177]

Shock

Shock can be defined as a “state in which a profound and widespread reduction of effective tissue perfusion leads to reversible, and, if prolonged, irreversible cellular injury.”[178] Based on hemodynamic profile, shock is classified into four categories: cardiogenic, hypovolemic, extracardiac obstructive, and distributive.[178]

Physiologic Effects of Shock

All forms of shock exhibit common cellular metabolic processes that typically lead to cell injury, organ failure, and eventually, death.[178,179] These processes are caused by multiple interrelated factors, including cellular ischemia, circulating or local inflammatory mediators, and free radical injury.[178,179]

Shock elicits at least three respiratory responses: increase in dead space ventilation, respiratory muscle dysfunction, and pulmonary inflammation. The increase in dead space ventilation is an early accompaniment of shock.[180–182] It results from a fall in pulmonary perfusion.[178] (The terminology of dead space is confusing. *Anatomic dead space* is made up of the conducting airways (nose, mouth, pharynx, larynx, trachea, bronchi, and bronchioles). *Alveolar dead space* is made up of alveoli that receive some or no blood flow, which does not match ventilation (units with very high $\dot{V}_A/\dot{Q}$ ratio). *Physiologic dead space* is the sum of anatomic dead space and alveolar dead space.) $\dot{V}_E$ increases to achieve normocapnia.[182,183] $\dot{V}_E$ also may further increase through other mechanisms. First, baroreflex deactivation (secondary to hypotension) amplifies the ventilatory response to stimulation of the peripheral chemoreceptors.[184] Second, direct stimulation of the aortic chemoreceptors occurs as a result of decreased oxygen delivery.[185] (This point, although clearly demonstrated in the cat,[185] remains controversial in humans.[75]) Third, cerebral hypoperfusion causes intracellular hypercapnia and acidosis[72] and, in turn, induces further hyperventilation unless the neurons are depressed (e.g., by hypoxia, lack of substrates, or excessive accumulation of metabolic products).[72] Fourth, increased respiratory drive occurs as a result of peripheral stimulation of pulmonary J receptors.[178] Fifth, increased respiratory drive results from vestibular activation during orthostasis (vestibulorespiratory reflex).[186] Sixth, $\dot{V}_E$ increases to compensate for lactic acidosis,[187] which results in part from the overworked and underperfused respiratory muscles.[182,188,189] The increase in $\dot{V}_E$ accompanying these responses may enhance venous return (through the respiratory pump) and vasoconstriction (through reduced Pa_{CO_2}), helping the cardiovascular system to cope with hypovolemia.[190]

Respiratory muscle dysfunction is one result of the associated cellular dysfunction and injury.[136,191] Many mechanisms contribute to this dysfunction in septic shock, including failure of neuromuscular transmission (because of elevated of muscle membrane potential and failure of excitation–contraction coupling), the cytotoxic effect of nitric oxide and its metabolites, free radicals, ubiquitin-proteasome proteolysis, and, possibly, a decrease in nicotinic acetylcholine receptors.[136,192] Local dysregulation of the circulation and of the Krebs cycle also may contribute.[136] Many pathways purported to be responsible for respiratory muscle dysfunction in sepsis are also activated in cardiogenic and hemorrhagic shock.[193–198]

Laboratory animals with cardiogenic[199] and septic shock[192,200] die of respiratory failure. Death is not caused by pulmonary disease per se but by an inability of the respiratory muscles to maintain adequate ventilation. In dogs with cardiogenic shock, institution of mechanical ventilation decreases the metabolic needs and thus the blood flow to the respiratory muscles (from 21% to 3% of the total cardiac output).[201] A nonrandomized study by Kontoyannis et al[202] in twenty-eight patients with cardiogenic shock provides support for the view that hemodynamic instability is an indication for mechanical ventilation. Compared with nonventilated patients, ventilated patients were weaned from an intraaortic balloon pump more often, and their survival was greater.[202]

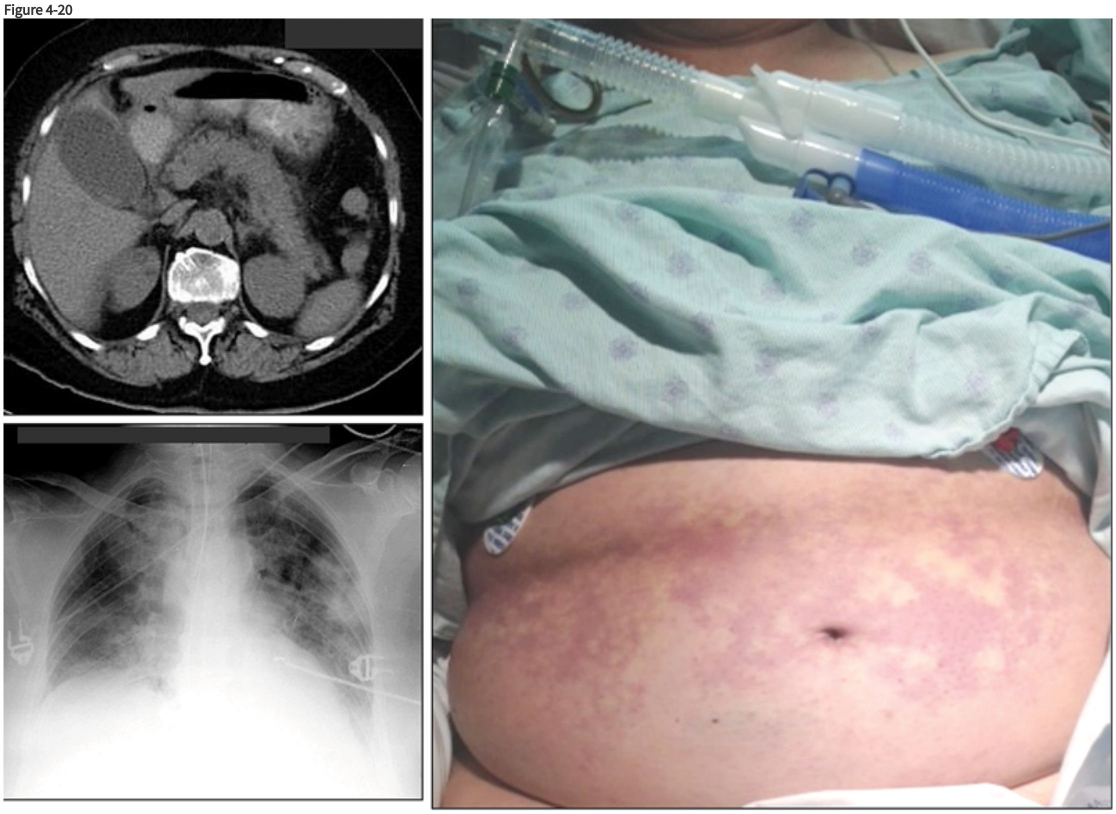
The failure to reverse shock and treat the underlying cause promptly predisposes to ARDS (see Chapter 29). Pulmonary vascular resistance increases during shock, including septic shock (in which peripheral vascular resistance is usually decreased).[203]

Hypovolemic, distributive, and extracardiac obstructive shock result in decreased diastolic filling. Low-pressure stretch receptors located in right atrium and pulmonary artery consequently signal the medullary vasomotor centers, triggering sympathetic discharge.[178,204] High-pressure baroreceptors in the aortic arch contribute (negative feedback to the tonic discharge of the medullary vasoconstrictor centers) to the vasomotor response of shock as long as the mean arterial pressure is no lower than 80 to 90 mm Hg.[205] When mean pressure is less than 80 to 90 mm Hg, the aortic baroreceptor response is eliminated (baroreceptor deactivation).[178,205] Likewise, the carotid baroreceptors contribute (negative feedback) to the vasomotor response as long as the mean pressure is no lower than 60 mm Hg.[205] When mean pressure is less than 60 mm Hg, the carotid baroreceptor response is eliminated.[205] When mean pressure is less than 50 to 60 mm Hg, the peripheral chemoreceptors (sensitive to Pa_{O_2}, Pa_{CO_2}, and pH) dominate.[178] The most powerful stimulus to sympathetic tone during severe shock, however, is the ischemic response of the CNS.[205] When mean pressure falls below 50 to 60 mm Hg, the medullary chemoreceptors become active.[205] Maximal sympathetic stimulation is induced by these receptors when mean pressure is 15 to 20 mm Hg, resulting in maximal cardiovascular stimulation.[205] The Cushing response to increased intracranial pressure is an example of this reflex operating in a different setting.[178] Increased sympathetic outflow from the CNS is aimed at supporting oxygen delivery to vital organs. Other compensatory responses include releases of adrenocorticotropic hormone, antidiuretic hormone, and aldosterone, which contribute to sodium retention and maintenance of cardiovascular catecholamine responsiveness.[178,179,206]

Clinical Presentation of Shock

Patients in shock or in the process of developing it may report dyspnea. Patients are usually tachypneic and tachycardic; they have primary respiratory alkalosis or a metabolic acidosis with some degree of respiratory compensation. Tachypnea, combined with low tidal volume, worsens dead space ventilation. The skin of patients in septic shock is initially warm and dry. It is typically cold and clammy when cardiac output is low. Livedo reticularis is sometimes seen in patients who have profound hypotension, necessitating the use of high-dose pressors (Fig. 4-20). Shock is usually not an all-or-none phenomenon that occurs abruptly after injury or infection. Instead, homeostatic compensatory mechanisms are engaged.[178,179] Early in the course, subtle signs of hemodynamic stress include tachycardia and decreased urine output. During early shock, assessment based on vital signs, central venous pressure, and urinary output may fail to detect global tissue hypoxia.[207] If the precipitating insult is too great or progresses quickly, compensatory mechanisms fail, and overt shock follows.[178]

Figure 4-20



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Livedo reticularis in shock. A 69-year-old woman developed severe gallstone pancreatitis *(left upper panel)* complicated by acute respiratory distress syndrome (ARDS) *(left lower panel)* and shock resistant to high-dose vasopressors. *Right:* A few hours after institution of mechanical ventilation the patient developed diffuse livedo reticularis of the skin, which is characterized by pale, ischemic, hypoperfused areas interspersed with erythematous areas of increased perfusion.

Compensatory mechanisms tend to protect the CNS from the ill effects of decreased cerebral perfusion. In the absence of cerebrovascular compromise, ischemic injury is unusual if mean arterial pressure is 50 to 60 mm Hg or higher.[178] Before ischemic injury, consciousness may become altered, depending on perfusion deficit. Contributory factors include electrolyte disturbances, hypoxemia, and hypercapnia.[178] Sepsis-related encephalopathy can occur at higher arterial pressures (secondary to inflammatory mediators); it is associated with increased mortality.[208] Altered consciousness by itself may be an indication for intubation.

Intubation versus Mechanical Ventilation

In some instances, patients require endotracheal intubation to maintain airway patency because of upper airway obstruction, an inability to protect the airway from aspiration, or to manage secretions. Not all intubated patients necessarily require ventilator support.

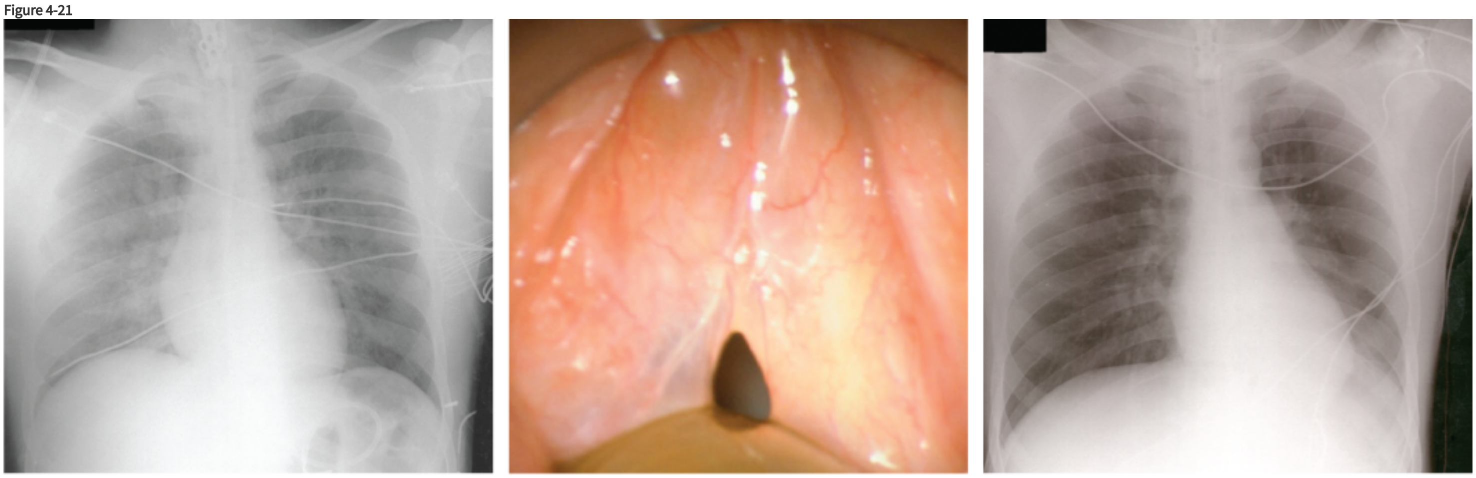
Upper Airway Obstruction

Upper airway obstruction is one of the most urgent and potentially lethal medical emergencies. Complete airway obstruction lasting for as little as 4 to 6 minutes can cause irreversible brain damage.[209] The upper airway, which encompasses the passage between the nares and carina,[210] can be obstructed for functional or anatomic reasons. Among the first are vocal cord paralysis and laryngospasm.[210–213] Among the second are trauma, burn, infections, foreign bodies, and tumors.[209,210] Functional and anatomic obstruction can occur postoperatively in patients with redundant pharyngeal soft tissue (sleep apnea) and loss of muscle tone related to postanesthetic state.[210–212]

The first warning of airway obstruction in an unconscious patient may be failure of a jaw-thrust maneuver to open the airway or an inability to ventilate with a bag valve.[209] In a conscious patient, respiratory distress, stridor, altered voice (aphonia or dysphonia), snoring, dysphagia, odynophagia, prominence of neck veins, and neck and facial swelling all may indicate impending airway obstruction.[209,210] Patients may bring their hands to their neck, a sign of choking.[210] Other signs include suprasternal and intercostal indrawing and reduced or absent air movement on auscultation. Wheezing may be present (or absent). Thoracoabdominal paradox may be prominent. Sympathetic discharge is high. Patients are diaphoretic, tachycardic, and hypertensive. As asphyxia progresses, bradycardia, hypotension, and death ensue.[210]

Upper airway obstruction can be complicated by pulmonary edema (Fig. 4-21)[210,211]—incidence of 11% in one adult series[214]—or pulmonary hemorrhage.[212] Increased venous return (more negative intrathoracic pressure and catecholamine-induced venoconstriction) contributes to pulmonary edema, but it cannot be the sole mechanism;[215] as intrathoracic pressure becomes more negative, venous return to the right ventricle becomes flow-limited.[216] Other factors contributing to pulmonary edema include decreased left-ventricular preload (leftward shift of interventricular septum), increased left-ventricular afterload (increased negative intrathoracic pressure and catecholamine-induced elevation of systemic vascular resistance), pulmonary vasoconstriction (hypoxemia and acidosis), and possibly, stress failure of the alveolar-capillary membrane.[212,217]

Figure 4-21



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Pulmonary edema caused by upper airway obstruction. *Left:* Portable anteroposterior chest radiograph of a 35-year-old man who developed pulmonary edema secondary to laryngeal edema. *Middle:* Laryngeal edema demonstrated by direct laryngoscopy. *Right:* Radiograph following resolution of pulmonary edema. (Courtesy of L. Baijens)

Whether pulmonary edema develops during (or after) relief of upper airway obstruction may depend on whether the obstruction is fixed or variable.[211] Fixed upper airway obstruction results in vigorous inspiratory efforts (Mueller maneuver) followed by vigorous expiratory efforts (Valsalva maneuver).[211,214] Exhalation against an obstructed airway raises intrathoracic and alveolar pressures. The positive expiratory pressure decreases pulmonary vascular filling and opposes the hydrostatic forces that favor transudation of fluid into the alveoli during inhalation.[214] With a sudden relief of obstruction, positive expiratory pressure is lost; consequently, there is a massive transudation of fluid from the pulmonary interstitium into the alveoli (pulmonary edema) over minutes to hours. In contrast to fixed obstruction, variable extrathoracic upper airway obstruction hinders inhalation. Exhalation usually is unaffected. In this situation, the hydrostatic forces, which favor transudation of fluid into the alveoli during inhalation, are unopposed, leading to edema before relief of the obstruction.[211]

Upper airway obstruction may worsen suddenly because resistance varies with the fourth power of the radius. A slight change in airway anatomy may increase resistive load dramatically.[210] For example, manipulation of the upper airway by an inexperienced clinician may induce edema, which can increase airway resistance markedly and induce asphyxia.

An initial assessment is undertaken to determine severity of airway compromise. If compromise is judged severe, the airway should be secured immediately. If ventilation is adequate, more detailed assessment is wise.[209] Arterial blood gases are not particularly helpful because they are not specific to airway patency.[209] They may show little change until a patient is in extremis.[209]

Steps to relieve airway obstruction include use of pharyngeal airways, endotracheal intubation, cricothyrotomy, tracheotomy, endoscopy, intubation over a fiber-optic bronchoscope, and medications (e.g., epinephrine, norepinephrine, antihistamines, steroids, and antibiotics).[218] In general, pharmacotherapy cannot reverse mechanical obstruction. Use of helium-oxygen mixtures should not engender a false sense of security.[218]

The nature of respiratory noises helps to localize lesions.[209] Stridor is an inspiratory sound typically caused by a lesion above the thoracic inlet (usually glottic or supraglottic). Wheezing is generated below this level.[209] Snoring, a feature of obstructive sleep apnea, can be life-threatening. A patient with an obstructed airway should not be sedated until the airway has been secured; minimal sedation may precipitate acute respiratory failure.

Cricothyrotomy and tracheotomy can be performed at the bedside or in the operating room. If time permits and the patient is conscious and moving sufficient air to speak, it may be best to transport the patient to the operating room.[218] Although percutaneous tracheotomy is gaining in popularity, it is best performed in an already intubated patient and not as an emergency procedure.[218]

Inability to Protect the Airway from Aspiration

Patients with severe bulbar weakness or decreased consciousness may be unable to protect the airway against aspiration.[219,220] The lack of a gag reflex or cough on suctioning suggests impaired protective reflexes.[221] If the cervical spine is stable, the head should be flexed while checking for airway obstruction. Inability to maintain a patent airway is an indication for elective intubation.[221]

Patients who require an oral airway or special appliances may require prophylactic intubation; such patients are unstable and can asphyxiate or vomit and aspirate suddenly.[221] Ideally, patients with depressed consciousness should be assessed while asleep, a time during which they are at greatest risk of obstruction.[221]

Severe head injury is a condition requiring intubation for airway protection. In the past, controlled hyperventilation (Pa_{CO_2} of 25 to 35 mm Hg) was delivered in these patients with the goal of reducing intracranial pressure. Such a strategy, however, has proven harmful and is no longer recommended.[222] It is unknown whether the use of short periods of hyperventilation to suddenly lower intracranial pressure are harmful or not.

Secretions

Occasionally, endotracheal intubation or tracheostomy may be required to manage a large amount of secretions or to remove secretions in severely debilitated patients who themselves cannot clear them.[59,144]

Many patients with the preceding conditions are capable of maintaining adequate gas exchange following endotracheal intubation. Yet most intensivists still connect such patients to a ventilator. This decision commonly is taken independently of any consideration about the work of breathing imposed by an endotracheal tube.[223]

*$A\text{-}aD_{O_2}$ is calculated as $PA_{O_2} - Pa_{O_2}$, where PA_{O_2} (alveolar O_2 tension) can be estimated according to the simplified alveolar gas equation:

$$PA_{O_2} = FI_{O_2} \times (P_B - PH_2O) - Pa_{CO_2}/R$$

where FI_{O_2} is fractional concentration of inspired O_2 (approximately 0.21 when breathing room air), P_B is barometric pressure (approximately 760 mm Hg at sea level), PH_2O is water vapor pressure (usually taken as 47 mm Hg at 37°C [98.6°F]), and *R* is respiratory exchange ratio of the whole lung. The respiratory exchange ratio (*R*) = CO_2 production/O_2 consumption ($\dot{V}_{CO_2}/\dot{V}_{O_2}$) is normally approximately 0.8. In steady state, *R* is determined by the relative proportions of free fatty acids, protein, and carbohydrate consumed by the tissues. In this equation, it is assumed that alveolar P_{CO_2} and Pa_{CO_2} are the same (usually they nearly are). In healthy young subjects (≤30 years old) breathing air at sea level, $A\text{-}aD_{O_2}$ is usually less than 10 mm Hg, but it increases to as much as 28 mm Hg in some healthy 60-year-old subjects.

Goals of Mechanical Ventilation

The fundamental goal of mechanical ventilation is to keep a patient alive and free from iatrogenic complications so that the catastrophic precipitating event(s) may resolve. Attention should be directed to the primary disorder.

Reversal of Apnea

The goal of mechanical ventilation in the apneic patient is to restore ventilation.

Reversal of Respiratory Distress

For obvious ethical reasons, no human studies have addressed the natural course of acute respiratory failure in the presence of increased work of breathing or, for that matter, with any other type of respiratory failure. Animal data indicate that increased loads can cause respiratory muscle damage, CO_2 retention, and as a terminal event respiratory muscle fatigue.[136] Increased load may be responsible for respiratory muscle damage in patients with COPD[224] and in patients dying while supported by mechanical ventilation.[225] In sepsis, increased respiratory efforts are particularly damaging to the respiratory muscles.[226]

Despite intense research, the role of contractile fatigue in the development of respiratory failure in patients is unknown.[136] Diaphragmatic contractility has been quantified objectively (phrenic nerve stimulation) in only one study where patients developed acute respiratory distress (during weaning from mechanical ventilation).[5] No change in diaphragmatic contractility was documented.[5] It is not known if the latter observation applies to patients in respiratory distress who have yet to undergo mechanical ventilation.

It seems self-evident that connecting a patient to a ventilator and providing ventilator assistance should unload the respiratory muscles and, possibly, reduce muscle stress. To date, however, we do not know the desirable level of unloading (and for how long) for a specific patient. Insufficient unloading can be dangerous to the respiratory muscles, as can excessive unloading (see Chapter 43).[227]

Although most patients in acute respiratory failure have increased work of breathing, this may not be the sole problem. Most patients also have abnormal gas exchange, impaired muscle perfusion, and sepsis-induced muscle dysfunction.[6,228,229] In patients with increased work of breathing, unloading by the ventilator may appreciably reduce $\dot{V}_{O_2}$ and $\dot{V}_{CO_2}$.[8,54] These reductions, in turn, may improve concurrent hypoxemia and hypercapnia.

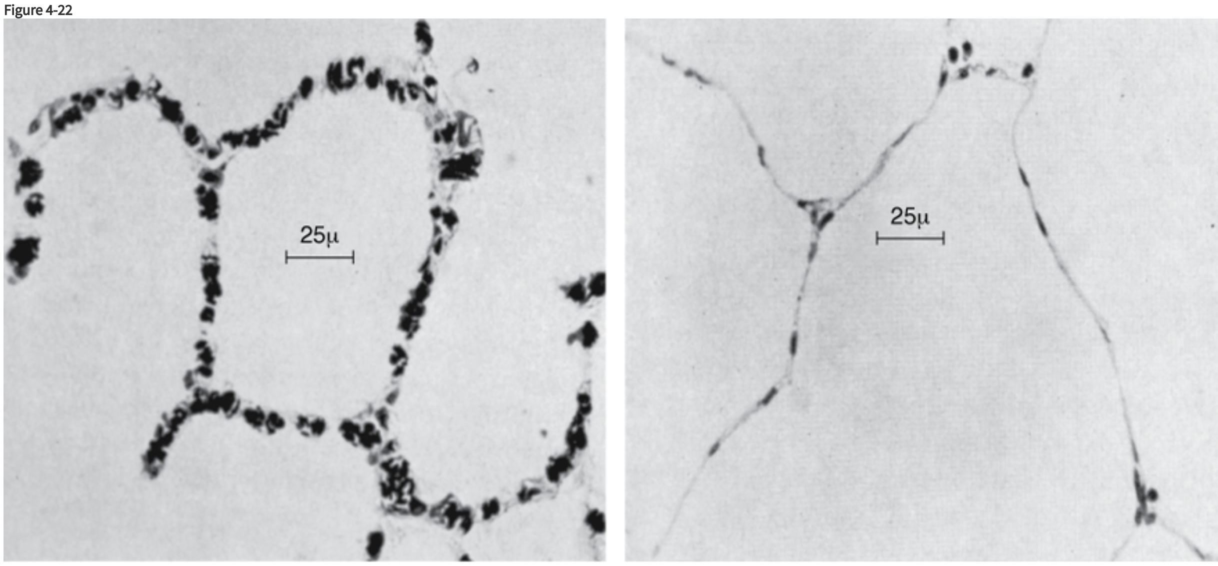
Reversal of Severe Hypoxemia

Mechanical ventilation is commonly commenced with 100% oxygen. The response helps to define the underlying pathophysiology and thus aids in differential diagnosis and therapy (see Figs. 4-9 and 4-12). For example, if 100% oxygen fails to increase Pa_{O_2} in a patient with an exacerbation of COPD, the underlying problem is not pure $\dot{V}_A/\dot{Q}$ mismatch (as is typical with acute bronchitis); instead, the patient has coexisting shunt. Common causes of shunt include pneumonia, congestive heart failure, atelectasis, and pulmonary embolism. (For a discussion of oxygen toxicity, see Chapter 45.)

Patients with increased shunt commonly exhibit considerable improvement in oxygenation with application of PEEP. The improvement results from a decrease in shunt[230] secondary to recruitment of previously atelectatic areas and redistribution of extravascular lung water from alveoli to peribronchial and perivascular spaces.[231] If cardiac output decreases (with PEEP), this can contribute to the decrease in shunt.[230,232]

PEEP causes an increase in dead space through several mechanisms. First, an increase in lung volume exerts radial traction on the airways, increasing their volume with a consequent increase in *anatomic dead space.* Second, increased airway pressure tends to divert blood flow from ventilated lung units by compressing capillaries. The consequent development of areas of high $\dot{V}_A/\dot{Q}$ ratio (or even unperfused areas) produces an increase in *alveolar dead space.* Such dead space is especially common in the uppermost lung units, where pulmonary artery pressure is relatively low because of the hydrostatic effect.[61,233] If the capillary pressure falls below airway pressure, the capillaries may collapse completely and the lung units become unperfused (Fig. 4-22).[233] Two factors encourage collapse: very high airway pressure and low venous return.

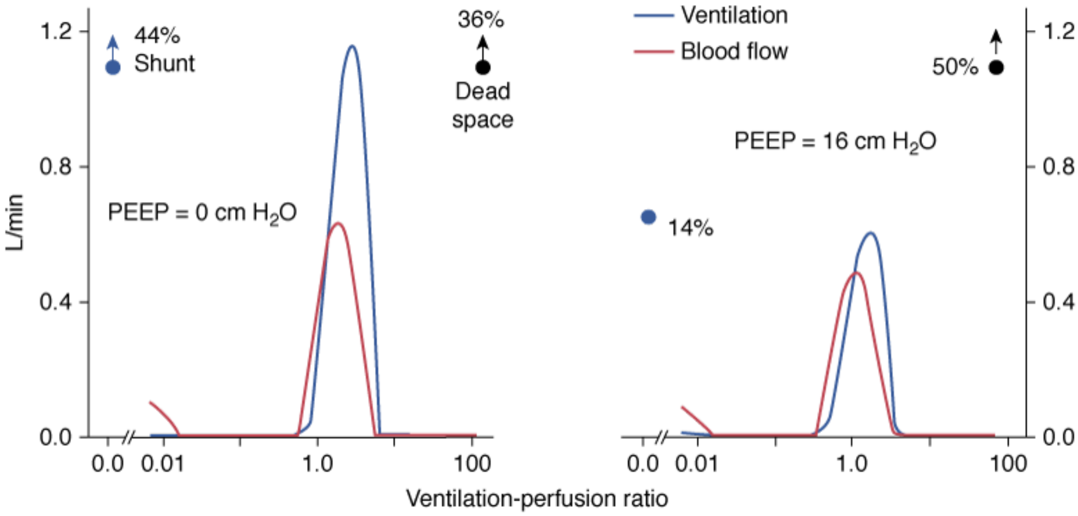
Figure 4-22



Effect of an elevated airway pressure on the structure of pulmonary capillaries. *Left:* Normal appearance. *Right:* An increase in alveolar pressure above capillary pressure produces capillary collapse. (Used, with permission, from Glazier et al.[233])

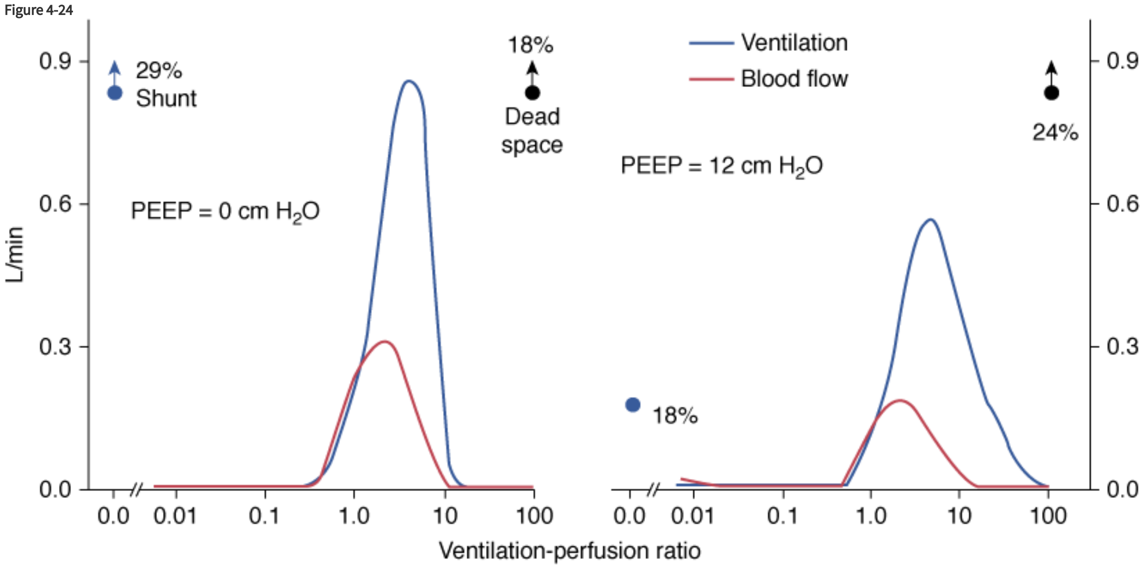
Dantzker et al[230] showed that increasing levels of PEEP can induce two distinct patterns of $\dot{V}_A/\dot{Q}$ distribution. Some patients experienced no change in the pattern of $\dot{V}_A/\dot{Q}$ relationships (Fig. 4-23). Other patients experienced broadening of the ventilation dispersion—increases of areas with high $\dot{V}_A/\dot{Q}$ ratios and alveolar dead space (Fig. 4-24). The improvement in gas exchange with PEEP is even greater when pressors are used to prevent the expected decrease in cardiac output during PEEP.[234]

Figure 4-23



Effect of positive end-expiratory pressure (PEEP) on shunt and dead space in a patient with acute respiratory distress syndrome. A progressive increase in PEEP from 0 to 16 cm H_2O caused a decrease in shunt from 44% to 14% of the cardiac output and an increase in dead space from 36% to 50% of the tidal volume. The shape of the distribution of ventilation and perfusion did not change. (Modified, with permission, from Dantzker et al.[230])

Figure 4-24



Effect of positive end-expiratory pressure (PEEP) on shunt and dead space in a patient with acute respiratory distress syndrome. An increase in PEEP from 0 to 12 cm H_2O produced a decrease in shunt (from 29% to 18%) and increase in dead space (from 18% to 24%). Also apparent is a widening of the dispersion in ventilation with the development of units with very high $\dot{V}_A/\dot{Q}$; these may have resulted from diversion of blood flow or increased ventilation of these areas of the lung. (Modified, with permission, from Dantzker et al.[230])

In some patients, PEEP causes no improvement or even a decrease in Pa_{CO_2}. This effect is the result of increased dead space ventilation, diversion of blood flow from well-ventilated to unventilated regions, and decreased cardiac output (especially if circulating blood volume is depleted).[61] Lack of improvement in oxygenation with PEEP also may result from a patent foramen ovale because PEEP can increases the right-to-left shunt.[235]

Because PEEP can decrease cardiac output,[230,236] its net effect should be judged in terms of oxygen delivery. Mixed venous P_{O_2}[207,236] has been used as a surrogate for oxygen delivery. Other potential hazards of PEEP include reduced splanchnic and renal blood flow, barotrauma, and ventilator-induced lung injury (see Chapter 10).

In addition to mechanical ventilation, patients with hypoxemic respiratory failure may require other therapies, such as antibiotics (pneumonia), diuretics, and inotropic support (heart failure). Ancillary strategies to reverse hypoxemia include exogenous surfactant, nitric oxide supplementation and prone position.[237] These strategies can increase oxygenation in patients with ARDS, but none has improved patient outcome.

Theoretically, patients with hypoxemia secondary to $\dot{V}_A/\dot{Q}$ mismatch can be managed by increasing F_{IO_2} without mechanical ventilation (see Fig. 4-12). In reality, such patients always have increased ventilatory requirements (see Fig. 4-10), which may demand ventilator support. Many patients with $\dot{V}_A/\dot{Q}$ inequality requiring mechanical ventilation are hyperinflated (e.g., COPD or status asthmaticus). Hyperinflation decreases the efficiency of the respiratory muscles in generating pressure, which contributes to the respiratory distress.[136]

In a study of seven patients with exacerbations of COPD, mechanical ventilation improved $\dot{V}_A/\dot{Q}$ mismatch by redistributing blood flow away from low $\dot{V}_A/\dot{Q}$ areas.[238] Dispersion of the distribution of ventilation also improved. Dead space or ventilation of high $\dot{V}_A/\dot{Q}$ units did not change.[238] Patients with increased $\dot{V}_A/\dot{Q}$ mismatch associated with COPD or ARDS may benefit from careful application of PEEP (see Chapter 10). Use of PEEP in patients with status asthmaticus is fraught with danger (see Chapter 30).

Reversal of Severe Hypercapnia

Severe hypercapnia depresses the CNS and decreases respiratory motor output.[239] A viscous cycle can arise: Hypercapnia depresses motor output leading to more hypercapnia.[240,241] Hypercapnia can decrease diaphragmatic contractility, although not consistently.[136] Acidosis may be more important than hypercapnia in depressing respiratory muscle contractility.[136]

The goal of mechanical ventilation in patients with hypercapnic respiratory failure is to improve $\dot{V}_A$. Ventilator strategies need to be tailored to the specific setting. In hypercapnic patients with status asthmaticus or COPD, the prolonged respiratory time constant poses a significant challenge. If the ventilator is set to deliver small tidal volumes at high respiratory frequencies, the prolonged time constant may interfere with lung emptying, and hyperinflation may ensue. Moreover, small tidal volumes may not achieve adequate alveolar ventilation because physiologic dead space is increased. Larger tidal volumes may achieve adequate alveolar ventilation but require longer expiratory times than do smaller tidal volumes. A common strategy to ensure sufficient time for exhalation is to increase inspiratory flow. The increase in flow decreases the time of mechanical inflation and, if respiratory rate remains constant, prolongs time available for exhalation. An increase in inspiratory flow, however, is commonly associated with an increase in respiratory rate.[242,243] Yet, despite the reduction in the respiratory cycle, the decrease in inspiratory time is accompanied by an increase in time available for exhalation—and a decrease in inspiratory effort.[243]

Neuromuscular disorders, such as Guillain-Barré syndrome, myasthenia gravis, and spinal cord injury, can cause hypercapnic respiratory failure.[136] These patients usually have normal lung mechanics, unlike patients with COPD or asthma. The normal time constant allows for greater flexibility in the setting of the ventilator.

Overzealous ventilation can cause serious complications, including life-threatening alkalosis, decreased cerebral perfusion, and cardiovascular instability. Patients previously hypercapnic are especially vulnerable to these complications. When severe, alkalosis is occasionally associated with coronary artery spasm,[244] confusion, myoclonus, asterixis, and seizures.[245]

Respiratory alkalosis reduces ionized calcium. For each 0.1-unit rise in pH, ionized calcium falls by 0.05 mmoL/L.[246] These changes are too modest and inconsistent[247] to account for the increased central and peripheral excitability associated with alkalosis. Paresthesias, carpal-pedal spasm, and tetany, seen in acute hyperventilation, are caused by the direct effects of respiratory alkalosis on neurons.[248] Other effects of alkalosis include increase in hemoglobin affinity for oxygen and, in the presence of increased shunt, a possible worsening of $\dot{V}_A/\dot{Q}$ relationship (secondary to a decrease in hypoxic pulmonary vasoconstriction).[249] Precipitous decrease in reduces blood flow to the CNS,[250] which may contribute to confusion and loss of consciousness in patients with hyperventilation.[245]

The most common hemodynamic instability associated with overzealous ventilator management of the hypercapnic patient (prolonged time constant) is hypotension. Hypotension often results from an increase in intrinsic PEEP after intubation—although a decrease in sympathetic tone caused by the decrease in Pa_{CO_2} and administration of sedation may be contributory factors. In this setting, the circulation usually is restored promptly by stopping ventilation for 30 seconds (or more) and then resuming less vigorous ventilation.

In the 1940s and 1950s, it was reported that rapid CO_2 washout after hypercapnia could cause hypotension (removal of cyclopropane anesthesia in humans)[251,252] and life-threatening ventricular arrhythmias (dog experiments).[253,254] Hyperkalemia appeared to be involved.[253,254] More recent series in patients, however, have not substantiated the earlier studies.[102,255] For example, Prys-Roberts et al[102,255] reported no electrocardiographic alterations when Pa_{CO_2} was reduced from approximately 80 mm Hg to less than 20 mm Hg over 5 minutes in anesthetized patients. Some practitioners suggest that supraventricular and ventricular arrhythmias associated with alkalemia[256] may occur only in patients with ischemic heart disease.[257] Whether reductions in ionized magnesium could contribute to cardiac irritability remains unclear.[247]

Excessive ventilation, over time, causes bicarbonate wasting by the kidney. In patients who retain CO_2 when clinically stable, this renal wasting (of bicarbonate) will increase ventilatory demands during weaning.

Goals of Mechanical Ventilation in Postoperative Respiratory Failure and Trauma

Patients developing postoperative hypoxemia usually are treated with supplemental oxygen and chest physical therapy (including incentive spirometry).[258–260] In approximately 10% of patients undergoing major elective abdominal surgery, supplemental oxygen and chest physical therapy do not prevent respiratory failure.[258] Squadrone et al[258] undertook a randomized study in more than 200 patients who had undergone major elective abdominal surgery and who developed hypoxemia within 1 hour of the operation. They compared the incidence of intubation in patients receiving standard treatment (50% oxygen and chest physical therapy) and patients who also received 7.5 cm H_2O CPAP (delivered noninvasively with a helmet). Compared with the control group, patients receiving CPAP had lower rates of intubation (1% vs. 10%) and complications (e.g., pneumonia, infection, and sepsis). The CPAP group spent fewer days in the ICU. Exclusion criteria included history of COPD, asthma, sleep apnea, heart failure, hypercapnia, and respiratory acidosis. Thus, these results[258] may not apply to those patients who are at greatest risk of postoperative atelectasis.

Bonde et al[149] undertook a prospective, randomized study in 102 patients undergoing elective lung surgery who were considered at high risk for retention of secretions.[144] Minitracheotomy (4-mm percutaneous cricothyroidotomy device) was performed at the conclusion of surgery in one group. Sputum retention was 30% in a conventionally treated group and 2% in the minitracheotomy group ($p < 0.005$).[149] Atelectasis was less common in the minitracheotomy group. Incidences of pneumonia, respiratory failure, myocardial infarction, and death were not affected by minitracheotomy.[149] Significant complications of minitracheotomy have been reported, however.[261,262] Therefore, its routine use, even in patients at risk of secretion retention, needs to be considered on a case-by-case basis.

Some patients with multiple trauma present with a flail chest. Many such patients may have respiratory failure secondary to underlying lung damage or other pathophysiology and may require mechanical ventilation. Flail chest on its own, however, is not an indication for mechanical ventilation.[263] In one randomized study of patients with flail chest who were hypoxemic and in respiratory distress, noninvasive CPAP decreased mortality and nosocomial infection as compared with patients who underwent endotracheal intubation and ventilation.[264]

Goals of Mechanical Ventilation in Shock

In hemodynamically unstable patients, tissue perfusion, including that of the CNS, is compromised.[179,207] Two main goals are to establish an adequate airway and reduce $\dot{V}_{O_2}$.[178] By resting the respiratory muscles and allowing for sedation, mechanical ventilation can reduce $\dot{V}_{O_2}$[265,266] and decrease sympathetic tone.[54] These effects may improve tissue perfusion,[201,267] which may explain why ventilator support improves outcome in animals[199] and patients in shock.[202] It is important to achieve good patient–ventilator synchronization;[268,269] otherwise, work of breathing increases, which diverts blood to the respiratory muscles and away from other vulnerable tissue beds.

Delivery of Mechanical Ventilation: Invasive versus Noninvasive Mechanical Ventilation

Solid experimental data support the use of noninvasive ventilation in patients with acute respiratory failure caused by severe exacerbations of COPD (limited to patients deemed not to require immediate intubation).[270] Noninvasive ventilation is probably superior to invasive ventilation in patients with cardiogenic pulmonary edema complicated with hypercapnia[271] and in immunocompromised patients with early hypoxemic respiratory failure.[270]

Sinuff et al[272] studied the use of a practice guideline for noninvasive ventilation in patients admitted to hospital for an exacerbation of COPD or congestive heart failure. The developers of the guideline pointed out that data from randomized trials do not support use of noninvasive ventilation in other disease states. Before introduction of the guideline, 65% of patients with diseases other than COPD or cardiogenic pulmonary edema were managed without endotracheal intubation. After introduction of the guideline, 100% of these patients were intubated; mortality also tended to increase (from 20.5% to 34.3%). This study raises the possibility that introduction of a practice guideline may cause an increase in morbidity and mortality.[272,273] A practice guideline could discourage physicians from the use of noninvasive ventilation in subgroups of patients who are likely to benefit from its use simply because benefit has not yet been demonstrated in a randomized trial and thus has not been incorporated into the practice guideline. In an accompanying editorial, Hill[273] commented: "The concern is that by classifying a sizable category of patients as 'not meeting noninvasive positive pressure ventilation criteria,' the authors could have unintentionally encouraged endotracheal intubation in this subgroup, possibly contributing to morbidity and mortality."

Chapters 18 and 34 provide detailed discussion of noninvasive ventilation.

Indications for Mechanical Ventilation and Nosology

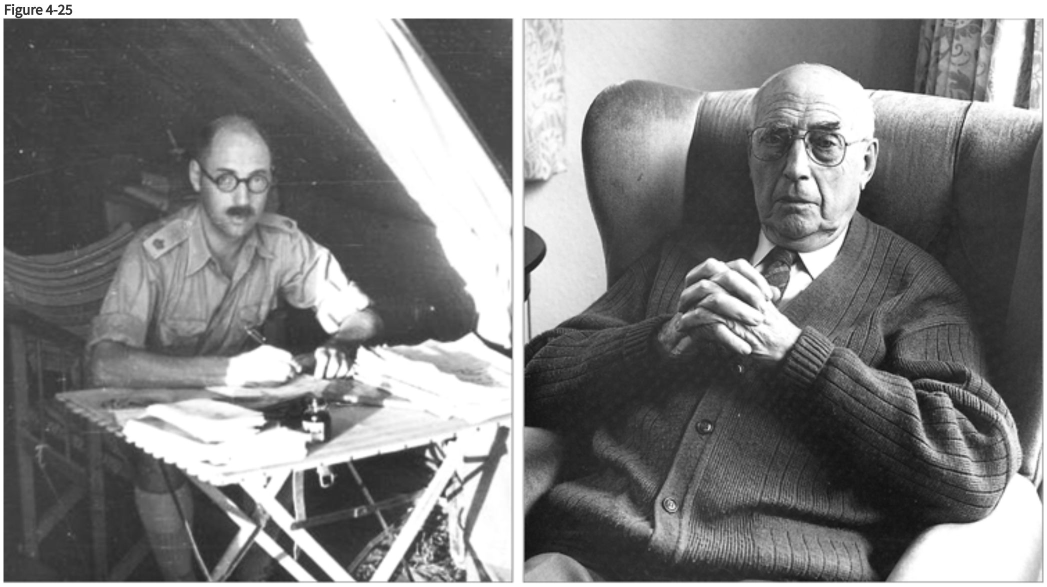
Indications: True versus Stated

In publications on mechanical ventilation, listed indications commonly include acute respiratory failure, exacerbation of chronic respiratory failure (secondary to infection, bronchoconstriction, heart failure), coma, and neuromuscular disease. Many patients, however, have these same conditions but do not require ventilator assistance. Consider COPD, one of the most common indications for mechanical ventilation. At any point in time, much, much less than 1% of patients with COPD are receiving mechanical ventilation. This small subset is commonly identified as those with acute respiratory failure. But the definition of acute respiratory failure is vague and the criteria loose. The usual defining criterion is a Pa_{O_2} of less than 60 mm Hg (sometimes combined with a Pa_{CO_2} of >50 mm Hg). It is patently absurd to suggest, however, that all patients with a Pa_{O_2} of 59 mm Hg (or lower) need ventilator assistance and that all patients with a Pa_{O_2} of 61 mm Hg (or higher) can be managed without it.

As such, the usually stated indications for mechanical ventilation are elastic, lacking meaningful boundaries. What, then, is the real reason to institute it? We believe that the most honest description of a physician's judgment at this juncture is: "The patient looks like he (or she) needs to be placed on the ventilator." That is, a physician institutes mechanical ventilation based on his or her gestalt of disease severity as opposed to slotting a patient into a particular diagnostic pigeonhole.

At first blush, this admission makes the undertaking of mechanical ventilation appear less scientific than other areas of medicine. Students of medicine are taught to make a diagnosis before initiating treatment. The usual teaching is that an accurate diagnosis will make the appropriate treatment relatively obvious. But this medical model is plausible only for diseases where the precise etiology is known (such as a microbial agent). The medical model is also more ideal than real. To understand the limitations of this model—and the apparent lack of science concerning ventilator indications—the reader needs to grapple with nosology, the discipline that names and classifies diseases. Guy Scadding (1907–1999)[274–276] wrote more lucidly on nosology than most; he made explicit the factual implications of medical usage of disease names. The account in this chapter borrows extensively from his writings (Fig. 4-25).

Figure 4-25



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Guy Scadding was one of the founders of the British Thoracic Society and first editor of *Thorax.* He served in the Royal Army Medical Corps during World War II *(left),* and is sometimes credited with changing the outcome of the war through his successful treatment of the pneumonia that Winston Churchill developed in Tunisia in December 1943.[274] Scadding wrote extensively on use of medical semantics and nosology. (Used, with permission, from Scadding[275] and Sinclair et al.[276])

Nosology

The first attempt to introduce a systematic and consistent nomenclature for diagnostic terms was made in the mid-nineteenth century.[277] Diseases were no longer viewed in terms of a galenic humoral disequilibrium. Instead, they were regarded as discrete entities—real things. This ontologic model grew out of the increasing use of autopsy, which was seen to uncover the true reasons for the corporeal changes induced by disease.[278] Ideally, each disease category would be identified through elicitation of pathognomonic signs on physical examination and the finding of a defining lesion on autopsy. This thinking is conveyed in the quip circulated by nineteenth-century physicians: If you were suffering from some mysterious illness, the best thing to do was go to Vienna (then the Mecca of medical science)[279] and be diagnosed by Skoda and autopsied by Rokitansky.

Disease Definition and Characteristics

How is disease defined? A *disease* is "the sum of abnormal phenomena displayed by a group of patients in association with a specified common characteristic (or set of characteristics) by which these patients differ from the norm (of healthy people) in such a way as to place them at a biological disadvantage."[280]

There are four main classes of characteristics by which diseases can be defined:

1. *Syndrome.* Historically, diseases are defined initially by way of a description of symptoms and signs; when these constitute a recognizable pattern, they are referred to as a *syndrome* (e.g., ARDS).

2. *Disorders of structure (morbid anatomy).* When a specifiable disorder is found to be associated constantly with a morbid-anatomic change, it tends to be named in these terms (e.g., the switch from jaundice to hepatitis).
3. *Disorders of function (pathophysiology).* When a disorder is found to be associated constantly with a specific abnormality of function, the abnormality may be used to name the disorder (e.g., hypothyroidism).
4. *Causation (etiology).* When the cause of a disease is discovered, the disease generally is redefined in causal terms (Legionnaire disease). Scadding[280] refers to another category, "clinical entity," that always needs an ad hoc explanation; he says, "It often seems to be the refuge of one who has not succeeded in clarifying his or her thoughts, but is nevertheless determined to put them into words."

In general, the direction of scientific advance follows the preceding sequence, although many conditions are never described in etiologic terms. The primary purpose of applying a name to a disorder is to provide a brief statement of the medical understanding of its nature (from syndrome to etiologic mechanism) and to serve as a verbal device for ease of communication. The American-European consensus definition of ARDS,[281] for example, has served as the basis of patient recruitment for most of the recent trials of mechanical ventilation in ARDS. Yet, as discussed in detail by Marini (see Chapter 31), this definition lacks scientific rigor. Nevertheless, in everyday practice, the term *ARDS* helps a clinician to predict prognosis and prescribe treatment. Moreover, in emergency settings (such as the ICU), problems are discussed and major decisions often are made without making any explicit diagnosis. Indeed, this is the rule rather than the exception when instituting mechanical ventilation. An experienced critical care physician can identify a patient who will die if left untreated but who might live if managed by mechanical ventilation—even though the physician is unable to identify the etiology of that patient's illness. Nevertheless, in such situations where physicians cannot put forward a defensible diagnosis, they still apply descriptors (at Scadding's level of "clinical entity"), such as "the patient is tiring out," to justify their judgment that mechanical ventilation is indicated.

Definitions: Essentialist and Nominalist

What are the factual implications of the naming of a disease? Diseases are defined in essentialist or nominalist terms. An *essentialist definition* tries to describe the true essence of an entity: the essential quality (invariable and fixed properties) that makes a given entity the type of thing it is—the "whatness" of an entity. (The study of the essence of things is called *ontology.*[282]) Essentialist ideas about diseases are implicit in everyday speech.[283] For example, a patient presents with a cough and mucoid sputum. The doctor makes a diagnosis of chronic bronchitis. The patient then thinks that chronic bronchitis is causing his or her cough. Given that chronic bronchitis is defined as a productive cough, the patient's reasoning is circular.

Such usage lays a linguistic trap: Many laypeople and some doctors think that the names of diseases refer to active agents that cause the illness. To talk of diseases as if they existed as real entities is plausible (at first sight) only in relation to diseases that are defined in etiologic terms.[284] Even then we must not confuse an etiologic agent with the disease itself. The disease is the effect on the affected person; diseases have no existence apart from patients. We treat patients, not diseases. When we speak of treatment of a disease, we are employing an ellipsis for treatment of patients with that disease.[280] All this brings to mind Osler's admonition: "It is much more important to know what sort of patient has the disease than to know what sort of disease the patient has."[278]

A *nominalist definition* recognizes that the task of revealing the essence of the *definiendum* is impossible.[285,286] Instead, it simply uses words to state the set of characteristics that are used to identify a member of a class (make a diagnosis). Such a definition makes it possible to determine whether a particular example (clinical picture) falls into a category to which a name (a disease) is applied.[287] A nominalist definition avoids the essentialist fallacy of regarding diseases as causes of illness; instead, it is simply naming a class of entities or events.

The nominalist–essentialist distinction becomes clearer if we consider the definition of acute respiratory failure. Karl Popper, who condemned essentialist definitions, observed that a good definition in science should be read from right to left, not left to right.[288] Consider the sentence, "Acute respiratory failure is the presence of a Pa_{O_2} of less than 60 mm Hg, with or without a Pa_{CO_2} of greater than 50 mm Hg, together with physical findings indicative of increased work of breathing." Reading from right to left, the sentence provides a "nominalist" answer to the clinician's question, "What shall we call the presence of a Pa_{O_2} of less than 60 mm Hg, with or without a Pa_{CO_2} of greater than 50 mm Hg, together with physical findings indicative of increased work of breathing?" rather than providing an "essentialist" answer to the question, "What is acute respiratory failure?" That is, the term *acute respiratory failure* is handy shorthand for the longer, more cumbersome description. Nothing more. The term *acute respiratory failure* contains no information about medicine, and nothing is to be gained from analyzing it.

Diagnostic Process, Treatment, and Value Judgment

The process of making a diagnosis goes through two broad steps. First, the physician undertakes an initial review of the clinical features, looking for a pattern that suggests one or more diseases. For example, a clinician notes eyelid retraction, tracheal tug, sternomastoid contraction, tachypnea, and monosyllabic speech. The physician concludes that the patient is in acute respiratory distress (an "entity" rather than a disease). Second, the physician undertakes a directed search for the defining characteristics (pathognomonic findings) of each of a number of suspected diseases.[289] Let us consider a patient who exhibits all the above-listed features of acute respiratory distress. On learning that the patient had been extubated a half hour previously, the physician suspects laryngeal edema and carefully listens for stridor. In a second patient, a physician's initial assessment again may reveal the general features of acute respiratory distress. On learning that this patient also has fever, chills, and rust-colored sputum, the physician suspects pneumonic consolidation. The physician then undertakes careful palpation (for tactile vocal fremitus), percussion (for dullness), and auscultation (for whispering pectoriloquy, egophony, and bronchial breathing).

The clinical diagnostic criteria are the descriptive features that best discriminate between one disease and other diseases with which it might be confused. In the best-case scenario, the clinical diagnostic criteria are made up largely of defining characteristics. None of the features is conclusive, but together they produce a degree of probability that justifies a diagnosis on which practical management may be based.[287] It is possible to state defining characteristics in objective, demonstrable terms when a disease is defined etiologically or as a disorder of structure or function.[280] The same is also possible for a disease defined syndromically if the description of the syndrome includes objectively demonstrable elements. For example, as entry criteria for a research study, respiratory distress might be defined (arbitrarily) as meeting three of the following four elements: respiratory rate greater than 33 breaths/min, a Pa_{O_2}/F_{IO_2} ratio of less than 300, phasic sternomastoid contraction, and nasal flaring. That is, in the context of this study, respiratory distress can be defined without making subjective value judgments.

When making decisions about the treatment of an individual patient, however, it is not possible to avoid subjective value judgments (things being assessed on a scale of goodness or badness). Ultimately, the decision of whether to institute mechanical ventilation (or not) boils down to a value judgment by the patient's physician. In some instances this decision will be preceded by a physician's making of a diagnosis. In many cases, however, physicians institute mechanical ventilation without having formulated a precise diagnosis. Along the same lines, Gross[290] has argued that it is unlikely that management of asthma would be improved were it possible to articulate a more widely accepted definition of this disease.

Factual Implications of Disease Terminology

Nosology is rarely discussed at medical conferences.[277] Questions on terminology are regarded as recondite and pedantic, eliciting yawns from the audience. When a speaker is asked to define the clinical entity about which he or she is speaking, the speaker may appear puzzled—believing that everyone surely knows what the term means. The audience becomes restless, seeing the question as a philosophical diversion that distracts from the hard scientific facts that the speaker is trying to discuss. Yet it makes little sense for a speaker to present detailed data analysis on a condition that the speaker cannot define. Likewise, readers should treat with a jaundiced eye statistics in surveys that list precise diagnoses for which mechanical ventilation was used. The ghost of such unrealistic (and unattainable) precision also hovers over lists of reasons for why patients were intubated in reports on controlled trials of noninvasive ventilation versus conventional therapy.

The application of precise mathematical methods to vague and ill-defined concepts gives them a false air of respectability that cloaks ignorance and perpetuates confusion.[284] It is unfortunate that the more fundamental the concept to which a word refers, the less careful we tend to be about the use of a clear definition.[291]

Contraindications to Mechanical Ventilation

Complications associated with mechanical ventilation can be lethal (see Chapters 43, 44, 45, 46, 47). Thus, mechanical ventilation should be used only when it is clearly needed. Intubation is not the first approach for most patients with an exacerbation of COPD; instead, noninvasive ventilation is the first choice. The same sequence probably holds for selected patients with congestive heart failure or immunocompromise. Mechanical ventilation should not be instituted when a mentally competent patient or a surrogate designated to make decisions on behalf of a noncompetent patient refuses it. If time permits, the patient and family should be instructed about the likely impact of mechanical ventilation on prognosis. For instance, hospital mortality of patients with idiopathic pulmonary fibrosis requiring mechanical ventilation is 68%[292] to 100%,[293] and 92% of the survivors are dead within 2 months of hospital discharge.[292]

Conclusion

When we started to write this chapter, we expected to end it by formulating a set of concrete recommendations as to when mechanical ventilation should be instituted. Readers willingly wade their way through complex pathophysiologic concepts if they believe the material enhances their understanding of a clinical topic. At the end, however, they expect to see the complexity reduced to a set of concrete recommendations, preferably conveyed as a list of entities with numerical values attached. That final step is not possible with this chapter. More than is the case for any other chapter in this book, it is not possible to articulate the indications for mechanical ventilation in the form of a list of items.

If it is not possible to formulate a list, then what? When you, dear reader, are in severe respiratory distress and a physician is standing at your bedside deciding whether or not to ventilate (and possibly intubate) you, what type of physician are you hoping will make this decision? We can speak only for ourselves. The physician we want is a person deeply versed in pathophysiologic concepts, skilled in the art of physical examination, with extensive experience of cases similar to our own illness, and blessed with good clinical judgment. We expect that physician to base the decision (on which our life depends) on his or her clinical gestalt. And we recognize that the physician may not be able to articulate the precise reasons behind this decision in the form of words.

Why can't our ideal physician express these thoughts in explicit terms? A wise physician standing at a patient's bedside senses a great deal of worthwhile information—much more than can be expressed in words. In short, there is a very large tacit coefficient to clinical knowledge—physicians *know* much more than they can communicate verbally.[294] There is an enormous difference between the assessment made by an experienced physician standing at a bedside and the assessment the same physician makes on hearing information (about the same patient) relayed over the telephone by a junior resident. An experienced and wise physician employs intuition rather than explicit rules in deciding what is best for a particular patient in a particular setting. A physician who regards such intuition as unscientific betrays a fundamental misunderstanding of the epistemology of science.[286]

Our failure to formulate a list of indications does not mean that we advocate a laissez-faire approach to instituting mechanical ventilation. Earlier we mentioned the absurdity of saying that mechanical ventilation is always indicated for acute respiratory failure, defined as a Pa_{O_2} of less than 60 mm Hg. This does not mean that we consider Pa_{O_2} unimportant. On learning that a patient has a sustained Pa_{O_2} of 40 mm Hg, a physician will take immediate steps to institute assisted ventilation. But it is not possible to pick a Pa_{O_2} breakpoint (between 40 and 60 mm Hg) below which the benefits of mechanical ventilation decidedly outweigh its hazards. It is futile to imagine that decision making about instituting mechanical ventilation can be condensed into an algorithm with numbers at each nodal point. In sum, an algorithm cannot replace the presence of a physician well skilled in the art of clinical evaluation who has a deep understanding of pathophysiologic principles.

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